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DUAL REFLECTION GROUPS OF GRADED ORE EXTENSIONS

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Abstract. We prove that the Artin–Schelter Gorenstein and Artin–Schelter regular properties descend from a graded Ore extension to its coefficient algebra, for every twisting automorphism and every skew derivation. We then determine the dual reflection groups of graded Ore extensions for all gradings compatible with the Ore structure. They are exactly the semidirect products of a dual reflection group of the coefficient algebra by the cyclic group generated by the degree of the new variable, meeting it trivially, and a nonzero skew derivation never contributes a new degree. For iterated Ore extensions we characterize the dual reflection groups by an order formula analogous to the theorem of Shephard, Todd and Chevalley, and we prove the binomial conjecture of Buzaglo and Rogalski for these algebras. We show that their covariant rings are complete intersections of growth type, whose Ext-algebras grow like a polynomial ring in as many variables as there are generators of nontrivial degree. This answers a question of Kirkman, Kuzmanovich and Zhang for these algebras. Finally, we describe the smash product as a Cayley quiver with relations, factor its matrix Hilbert series, and show that the extension becomes Hopf–Galois once the Jacobian is inverted.

1. Introduction

Let $\Bbbk$ be a field of characteristic zero, let V be a finite-dimensional $\Bbbk$-vector space, and let $W \subseteq GL(V)$ be a finite group acting on the polynomial ring $\Bbbk[V]$. The theorem of Shephard, Todd and Chevalley [41, 12] states that the invariant ring $\Bbbk[V]^W$ is again a polynomial ring if and only if W is generated by pseudo-reflections. In this case the invariant ring is generated by homogeneous elements $u_1, \dots, u_n$ of degrees $d_1, \dots, d_n$, and

$$|W| = d_1 d_2 \cdots d_n, \quad H_{\Bbbk[V]^W}(s) = \prod_{i=1}^{n} \frac{1}{1 - s^{d_i}}, \tag{1.1}$$

where H denotes the Hilbert series. Stanley described the relative invariants of such a group [43, Theorems 2.3 and 3.1]. For every linear character χ of W, the space of χ-relative invariants is a free $\Bbbk[V]^W$-module of rank one,

$$\Bbbk[V]^W_\chi = f_\chi \, \Bbbk[V]^W, \tag{1.2}$$

generated by a product f_χ of linear forms defining reflecting hyperplanes. The Jacobian $\det(\partial u_j / \partial x_i)$ is a relative invariant. Its character is the inverse of the determinant of the action of W on the linear forms, and up to a nonzero scalar it is the corresponding generator f_χ. The group W acts freely on the complement of its reflecting hyperplanes [44, Theorem 1.5]. Geometrically, the quotient map $V \to V/W$ is flat, its fibre over the origin is the spectrum of the coinvariant algebra $\Bbbk[V]/(u_1, \dots, u_n)$, which is a complete intersection, and W acts freely away from the reflecting hyperplanes. The results of this paper are noncommutative analogues of these three facts.

Artin–Schelter (AS) regular algebras are the noncommutative analogues of polynomial rings [3]. Their invariant theory studies actions of finite groups, and more generally of semisimple Hopf algebras, on AS regular algebras. In this setting the determinant is replaced by the homological determinant of Jørgensen and Zhang [22], and the theorem of Shephard, Todd and Chevalley has analogues for skew polynomial rings [24]. Kirkman, Kuzmanovich and Zhang studied the dual situation, in which a group algebra $\Bbbk G$ coacts on an AS regular algebra B [27]. A coaction of $\Bbbk G$ on B is the same as a G-grading

$$B = \bigoplus_{g \in G} B_g, \quad B_g B_h \subseteq B_{gh},$$

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and its coinvariants form the identity component B_e. Let B be a noetherian AS regular domain generated in degree one. A finite group G is a *dual reflection group* for B if B has a faithful G-grading, refining its $\mathbb{N}$-grading, whose identity component B_e is again AS regular [27, Definition 0.1].

The structure theory of reflection groups extends to dual reflection groups. If G is a dual reflection group for B, every component B_g is a free B_e-module of rank one on both sides,

$$B_g = f_g B_e = B_e f_g, \tag{1.3}$$

which is the analogue of (1.2). There is a unique element $m_G \in G$, the *mass element*, at which $\deg f_g$ is maximal, and f_{m_G} is the Jacobian of the coaction in the sense of Kirkman and Zhang [23, Definition 2.1]. The *covariant ring* $C(B) = B/B(B_e)_{\geq 1}$, where $(B_e)_{\geq 1}$ is the augmentation ideal of B_e, is a Frobenius algebra of dimension $|G|$, with basis the images of the elements f_g, and its Hilbert series is a palindromic product of cyclotomic polynomials [27, Theorems 0.3, 0.4 and 3.5]. In the commutative case the covariant ring is a complete intersection, and Kirkman, Kuzmanovich and Zhang asked whether $C(B)$ is always a complete intersection of growth type, in the sense that its Ext-algebra has finite Gelfand–Kirillov dimension [27, Question 0.7]. They noted that the question is open even for coactions of group algebras.

In low dimension, dual reflection groups are known through classification. In dimension two, Crawford classified the gradings of AS regular algebras by nonabelian finite groups and determined when the identity component is AS regular [16]. Buzaglo and Rogalski classified the coactions of cocommutative Hopf algebras on one-parameter skew polynomial rings in two variables, and in three variables for most values of the parameter [10], and the nonabelian dual reflection groups of three-dimensional AS regular algebras [11, Theorem 1.2]. They conjectured that an AS regular algebra with a dual reflection group whose identity component has no elements of degree one can be presented by binomial relations [11, Conjecture 1.4], and they proved it for nonabelian gradings of three-dimensional AS regular algebras [11, Corollary 5.15]. In dimension four, Goetz, Kirkman, Moore and Vashaw constructed families of dual reflection groups, among them groups of wreath product type coacting on skew polynomial extensions [19, Examples 2.1.6 and 2.1.7]. More recently, Crawford constructed dual reflection groups from solutions to the Yang–Baxter equation [17].

Ore extensions are noncommutative polynomial rings in one variable [36]. If A is AS regular and σ is a graded automorphism of A, then every graded Ore extension $B = A[t; \sigma, \delta]$ is again AS regular by the graded version of [30, Theorem 3.3]. Many of the algebras in the classifications above are iterated Ore extensions, and one wants to obtain the dual reflection groups of B from those of A. When $\delta = 0$, Buzaglo and Rogalski proved that a dual reflection group of B for which A and t are homogeneous is a semidirect product built from A [11, Proposition 6.1]. In dimension three, the Ore extensions with $\delta \neq 0$ are absent from their list of algebras with nonabelian dual reflection groups [11, Theorem 1.2]. This absence is established through the classification, and L. Buzaglo asked whether there is a structural reason behind it. This paper gives such a reason for gradings compatible with the Ore structure. We then classify the dual reflection groups of iterated Ore extensions with homogeneous Ore variables. We describe their covariant rings and determine the bigraded Poincaré series and the growth of the Ext-algebras of these rings. We also compute the associated McKay quivers and localizations.

Throughout, $\Bbbk$ is a field. In the structure theory of dual reflection groups recalled at the end of Section 2, and from Section 4 on, $\Bbbk$ is algebraically closed of characteristic zero. Gradings by groups are faithful and refine the $\mathbb{N}$-grading. If $b \in B_g$ is nonzero for a G-grading of B, we call g the *G-degree* of b. The *support* $\operatorname{Supp}(B)$ is the set of G-degrees of nonzero homogeneous elements of B. The symbol deg always denotes the $\mathbb{N}$-degree. The identity element of a group is e, and the order of an element γ is $|\gamma|$. We write $[r]_s = 1 + s + \cdots + s^{r-1}$.

Our first result is a descent theorem for the Gorenstein property. Let R be a noetherian connected graded algebra, let σ be a graded automorphism of R, and let δ be a σ-derivation of R such that

$$\delta(R_n) \subseteq R_{n+m} \quad \text{for all } n \geq 0, \tag{1.4}$$

for a fixed integer $m \geq 1$. The Ore extension $S = R[t; \sigma, \delta]$ is then connected graded with $\deg t = m$. The AS Gorenstein algebras and their Gorenstein parameters are recalled in Definition 2.1.

Theorem A. *Suppose that $S = R[t; \sigma, \delta]$ is AS Gorenstein of dimension D with Gorenstein parameter ℓ. Then R is AS Gorenstein of dimension $D - 1$ with Gorenstein parameter $\ell - m$. If moreover S is AS regular, then R is AS regular.*

Together with the preservation theorem of Liu, Wang and Wu, Theorem A shows that S is AS regular if and only if R is AS regular. Zhang and Zhang showed, in a setting that includes graded Ore extensions, that the Gorenstein condition passes from R to S [48, Proposition 3.2], and Theorem A gives the converse. When $\delta = 0$, the variable t is a normal regular element and $R \cong S/tS$, so the Gorenstein part of the theorem follows from the Rees lemma [28, Proposition 3.4(b)], and the regular part also uses the descent of the global dimension [33, §7.5]. The new case is $\delta \neq 0$. Then t is in general not normal, as for the Jordan plane, and the finiteness of the global dimension and the Hilbert series still descend by elementary arguments, but the Gorenstein condition requires another argument, which we obtain from local duality for S. We are not aware of a previous statement of this descent for $\delta \neq 0$, either in the Gorenstein or in the regular case. Theorem A is the Gorenstein counterpart of Phan's theorem that Koszulity descends along graded Ore extensions with $\deg t = 1$ [37, Corollary 1.3]. It holds for every degree m of the new variable and for AS Gorenstein algebras that are not regular.

We fix the setting for dual reflection groups. Let A be a noetherian AS regular domain generated in degree one, let σ be a graded automorphism of A, let δ be a σ-derivation of A with $\delta(A_n) \subseteq A_{n+1}$ for all n, and put

$$B = A[t; \sigma, \delta], \quad \deg t = 1.$$

A grading of B by a group G is *compatible with the Ore structure* if A is a graded subalgebra and t is homogeneous. We studied gradings of Ore extensions by arbitrary groups in [1] for differential polynomial rings and in [2] for general Ore extensions. By [2, Theorem A], a G-grading of A extends to a grading of B in which t is homogeneous of G-degree γ if and only if

$$\sigma(A_g) \subseteq A_{\gamma g \gamma^{-1}} \quad \text{and} \quad \delta(A_g) \subseteq A_{\gamma g} \quad \text{for all } g \in G.$$

When $\sigma = \mathrm{id}$, the degree γ commutes with the support of A [2, Remark 2.2(2)]. For general σ this fails, and this is the source of the semidirect products below. Our second result determines the dual reflection groups of B.

Theorem B. *Let G be a finite group with a grading of $B = A[t; \sigma, \delta]$ compatible with the Ore structure. Let $\gamma \in G$ be the G-degree of t, let $K = \mathrm{Supp}(A)$ and let $r = |\gamma|$. Then G is a dual reflection group for B if and only if K is a dual reflection group for A, for the grading of A by K, and*

$$K \cap \langle \gamma \rangle = \{e\}.$$

In this case $G = K \rtimes \langle \gamma \rangle$, the identity component is the graded Ore extension

$$B_e = A_e[t^r; \sigma^r|_{A_e}, \delta|_{A_e}],$$

where $\delta = 0$ whenever $r > 1$.

If G is a dual reflection group for B and $\delta \neq 0$, then $\gamma = e$. In this case σ and δ preserve every component A_g, and $(A_e)_1 \neq 0$.

When $\delta = 0$, Theorem B is [11, Proposition 6.1]. The case $\delta \neq 0$ is new, and the criterion is the same for every skew derivation. In particular a nonzero skew derivation never contributes a new G-degree to a dual reflection group, and the identity component of a dual reflection group of an Ore extension with $\delta \neq 0$ always has elements of degree one. If moreover $\dim A = 2$, then A_1 is spanned by two homogeneous elements, one of which lies in A_e, so $G = K$ is cyclic. This explains the absence of Ore extensions with $\delta \neq 0$ observed by Buzaglo and Rogalski in dimension three. Theorem B also explains the wreath product examples of Goetz, Kirkman, Moore and Vashaw (Example 5.13).

Applying Theorem B to each step of an iterated Ore extension determines the group completely. Let

$$B = \Bbbk[x_1][x_2; \sigma_2, \delta_2] \cdots [x_n; \sigma_n, \delta_n], \quad \deg x_i = 1, \tag{1.5}$$

where σ_i is a graded automorphism of the subalgebra $B^{(i-1)}$ generated by $x_1, \ldots, x_{i-1}$ and δ_i is a σ_i-derivation of $B^{(i-1)}$ raising degrees by one. Every grading of B in which each x_i is homogeneous is compatible with each step of (1.5). For such a grading we write $c_i \in G$ for the G-degree of x_i, and $r_i = |c_i|$.

Theorem C. *Let G be a finite group with a grading of the iterated Ore extension B of (1.5) in which every x_i is homogeneous. Then $G = \langle c_1 \rangle \langle c_2 \rangle \cdots \langle c_n \rangle$, and the following are equivalent.*

1. *G is a dual reflection group for B.*
2. *$|G| = |c_1|\,|c_2| \cdots |c_n|$.*
3. *Every $g \in G$ has a unique expression $g = c_1^{a_1} c_2^{a_2} \cdots c_n^{a_n}$ with $0 \leq a_i < |c_i|$.*

In this case G is solvable, $c_i = e$ whenever $\delta_i \neq 0$, and

$$f_{c_1^{a_1}\cdots c_n^{a_n}} = x_1^{a_1}\cdots x_n^{a_n} \quad (0 \le a_i < r_i),$$
$$f_{m_G} = x_1^{r_1-1}\cdots x_n^{r_n-1},$$
$$H_{B_e}(s) = \prod_{i=1}^{n} \frac{1}{1-s^{r_i}}.$$

If moreover $(B_e)_1 = 0$, then every δ_i is zero, the degrees $c_1, \dots, c_n$ are pairwise distinct, and B is presented by the binomial relations

$$x_i x_j = \lambda_{ij}\, x_{\pi_i(j)} x_i \quad (1 \le j < i \le n),$$

where $\lambda_{ij} \in \Bbbk^\times$, π_i is a permutation of $\{1, \dots, i-1\}$ and $c_{\pi_i(j)} = c_i c_j c_i^{-1}$.

Condition (2) is the analogue of $|W| = d_1 \cdots d_n$ in (1.1), with the orders r_i playing the role of the degrees of the basic invariants, and the formula for H_{B_e} is the analogue of the Hilbert series of $\Bbbk[V]^W$. The last part proves the conjecture of Buzaglo and Rogalski [11, Conjecture 1.4] for iterated Ore extensions with homogeneous Ore variables, in every dimension and for every finite group. The identity component is again an iterated Ore extension, on the generators $x_i^{r_i}$ (Proposition 5.6).

Condition (2) reduces the search for dual reflection groups to the finite quotients of an explicit finitely presented group. For instance, the dual reflection groups of $\Bbbk[x_1, \dots, x_n][t; \sigma]$, with σ a cyclic permutation of the variables, for gradings in which $x_1, \dots, x_n$ and t are homogeneous, are exactly the cyclic groups and the groups $\mathbb{Z}_r^n \rtimes \mathbb{Z}_{np}$ of Goetz, Kirkman, Moore and Vashaw (Corollary 5.16).

Our fourth result determines the bigraded Poincaré series of the covariant ring and the growth of its Ext-algebra. For a connected graded algebra C we write $E(C) = \bigoplus_{n\ge 0} \operatorname{Ext}^n_C(\Bbbk, \Bbbk)$ for its Ext-algebra and

$$P_C(y, s) = \sum_{n,j \ge 0} \dim_\Bbbk \operatorname{Ext}^n_C(\Bbbk, \Bbbk)_{-j}\; y^n s^j$$

for its bigraded Poincaré series. We write $\operatorname{gci}(C) = \operatorname{GKdim} E(C)$, so that C is a complete intersection of growth type exactly when $\operatorname{gci}(C)$ is finite [26, Definition 1.5].

Theorem D. *In the setting of Theorem C, suppose that G is a dual reflection group for B. Then $C(B)$ is an iterated truncated skew polynomial ring*

$$C(B) \cong \Bbbk[\bar{x}_{i_1}]/(\bar{x}_{i_1}^{r_{i_1}})\,[\bar{x}_{i_2}; \bar{\sigma}_{i_2}]/(\bar{x}_{i_2}^{r_{i_2}}) \cdots [\bar{x}_{i_m}; \bar{\sigma}_{i_m}]/(\bar{x}_{i_m}^{r_{i_m}}),$$

where $i_1 < \cdots < i_m$ are the indices with $c_i \neq e$ and $\bar{\sigma}_i$ is the automorphism induced by σ_i on the covariant ring of $B^{(i-1)}$. Its Poincaré series is

$$P_{C(B)}(y, s) = \prod_{c_i \neq e} \frac{1 + ys}{1 - y^2 s^{r_i}}.$$

In particular $C(B)$ is a complete intersection of growth type, and

$$\operatorname{gci}\big(C(B)\big) = \#\{i \mid c_i \neq e\}.$$

Theorem D answers [27, Question 0.7] positively for iterated Ore extensions with homogeneous Ore variables, for abelian and nonabelian groups alike. The Poincaré series coincides with that of the commutative complete intersection $\bigotimes_i \Bbbk[x_i]/(x_i^{r_i})$, the covariant ring of the diagonal reflection group $\prod_i \mu_{r_i}$, where μ_r is the group of r-th roots of unity. The proof uses an explicit minimal resolution of the trivial module over a truncated skew polynomial ring (Proposition 6.4).

Our fifth result concerns the smash products and McKay quivers of these gradings. A G-grading of B is an action of the dual group algebra $\Bbbk^G$, and the smash product $\Lambda = B\#\Bbbk^G$ is the analogue of the skew group algebra of a group action. It has orthogonal idempotents e_g indexed by G, with $e_g \Lambda e_h \cong B_{gh^{-1}}$ [14, Corollary 1.5]. For $c \in G$ let Π_c be the permutation matrix of left multiplication by c on G. The *Cayley quiver* of G with respect to $c_1, \dots, c_n$ has vertex set G and an arrow $h \to c_i h$ for every $h \in G$ and every i. The matrix Hilbert series of Λ is the matrix $H_\Lambda(s)$ of the Hilbert series of the spaces $e_g \Lambda e_h$, and $P_G(s)$ is the matrix with entries $s^{\deg f_{gh^{-1}}}$.

Theorem E. *Let G be a finite group with a grading of the iterated Ore extension B of (1.5) in which every x_i is homogeneous, and let $\Lambda = B\#\Bbbk^G$.*

(1) *The algebra Λ is the path algebra of the Cayley quiver of G with respect to $c_1, \dots, c_n$, modulo the lifts of the relations of B, and*

$$H_\Lambda(s) = (I - s\Pi_{c_1})^{-1}(I - s\Pi_{c_2})^{-1} \cdots (I - s\Pi_{c_n})^{-1}.$$

(2) *If G is a dual reflection group for B, then $H_\Lambda(s) = H_{B_e}(s)\, P_G(s)$ and*

$$P_G(s) = \prod_{i=1}^{n} \left(I + s\Pi_{c_i} + s^2\Pi_{c_i}^2 + \cdots + s^{r_i-1}\Pi_{c_i}^{r_i-1}\right).$$

The algebra $C(B)\#\Bbbk^G$ is Frobenius of dimension $|G|^2$, and the socle of its left projective module $\bar{\Lambda}e_h$ at the vertex h is the simple module at $m_G h$.

(3) *If G is a dual reflection group for B and $L = B[f_{m_G}^{-1}]$, then L is a crossed product of L_e by G, and*

$$L\#\Bbbk^G \cong M_{|G|}(L_e).$$

The factorization of P_G is the matrix form of the identity $H_{C(B)}(s) = \prod_i [r_i]_s$, and part (3) is the analogue of the fact that a reflection group acts freely off its reflecting hyperplanes. The description of Λ by a quiver with relations is an instance of covering theory [20, 13, 6] and of Crawford's description of smash products of Koszul AS regular algebras [15, Theorem 1.2]. For iterated Ore extensions the relations are the lifts of the Ore relations, and the matrix Hilbert series factors through the generators, also for the nonabelian groups of Theorem C. For the grading of $\Bbbk[x_1, x_2][t; \sigma]$ by the dihedral group of order eight, the quiver is the twisted cube of Figure 3. If G is abelian and $B = \Bbbk[V]$, then Λ is the skew group algebra of a diagonal reflection group. When this group is generated by reflections of order two, the quotient of Λ by the ideal generated by e_e was studied by Buchweitz, Faber and Ingalls [9, Theorem 4.17 and Corollary 4.19].

These results have a geometric meaning, which we explain in Section 8. In the language of noncommutative algebraic geometry [5], we regard B and B_e as algebraic analogues of the coordinate rings of an affine space X and of its quotient X/G. Then Theorem C is a Chevalley–Shephard–Todd theorem for iterated Ore extensions. The extension $B_e \subseteq B$ is free of rank $|G|$, and Theorem D says that its base change at the augmentation of B_e is a complete intersection of growth type. After inverting the Jacobian the extension becomes Hopf–Galois, and for iterated Ore extensions this localization is obtained by inverting the generators of nontrivial G-degree (Proposition 8.1). For the dihedral group of order eight, the point scheme of B is a triangle of lines, whereas B_e is a commutative polynomial ring (Example 8.4).

We also obtain formulas for homological determinants. If R is noetherian and AS regular and θ is a graded automorphism of $S = R[t; \sigma, \delta]$ with $\theta(R) = R$, then $\theta(t) = \lambda t + a$ with $\lambda \in \Bbbk^\times$, and

$$\operatorname{hdet}_S(\theta) = \lambda \operatorname{hdet}_R(\theta|_R).$$

When $\delta = 0$ and $a = 0$ this follows from a result of Jing and Zhang on normal elements [21, Proposition 2.4]. For a dual reflection group and a graded automorphism θ of A preserving every component, we prove

$$\operatorname{hdet}_{A_e}(\theta|_{A_e}) = \lambda_{m_G} \operatorname{hdet}_A(\theta), \quad \text{where } \theta(f_{m_G}) = \lambda_{m_G} f_{m_G}.$$

For iterated Ore extensions both determinants become explicit products.

All results on dual reflection groups assume that the grading is compatible with the Ore structure. Once A is a graded subalgebra, a homogeneous Ore variable can always be chosen after an affine change $t' = \lambda t + a$, at the cost of changing δ by an inner σ-derivation (Proposition 4.16). For the three-dimensional AS regular algebras of Ore type graded by nonabelian groups, the whole assumption holds up to a graded change of variables by the classification of Buzaglo and Rogalski [11, Theorem 3.4]. In dimension four, the dual reflection group of order sixteen of [19, §3.1] acts on an algebra whose presentation as a double Ore extension is not homogeneous for the grading, so this example lies outside the scope of the present methods (Remark 4.19).

Organization of the paper. Section 2 recalls AS regular algebras, graded Ore extensions, group gradings and the structure theory of dual reflection groups. Theorems A to E are proved in Sections 3 to 7, respectively. Section 8 interprets these results in terms of noncommutative quotients, and Section 9 contains the formulas for homological determinants.

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2. Preliminaries

In this section we fix the homological conventions and recall the structure theory of graded Ore extensions and dual reflection groups.

A $\Bbbk$-algebra R is *connected graded* if $R = \bigoplus_{n\geq 0} R_n$ with $R_i R_j \subseteq R_{i+j}$, $R_0 = \Bbbk$ and $\dim_\Bbbk R_n < \infty$ for all n. We write $R_+ = \bigoplus_{n>0} R_n$ for the augmentation ideal and $\Bbbk = R/R_+$ for the trivial module, on the left or on the right. Let $M = \bigoplus_{n\in\mathbb{Z}} M_n$ be a graded right R-module. For $j \in \mathbb{Z}$, the *shift* $M(j)$ is the graded module with

$$M(j)_n = M_{j+n}.$$

If M is bounded below and $\dim M_n < \infty$ for all n, its *Hilbert series* is the Laurent series

$$H_M(s) = \sum_{n\in\mathbb{Z}} (\dim_\Bbbk M_n)\, s^n,$$

so that $H_{M(j)}(s) = s^{-j} H_M(s)$. For graded modules M and N we put

$$\operatorname{Hom}_R(M,N) = \bigoplus_{j\in\mathbb{Z}} \operatorname{Hom}_{\mathrm{gr}}\big(M, N(j)\big),$$

where $\operatorname{Hom}_{\mathrm{gr}}$ denotes the degree-preserving homomorphisms, and $\operatorname{Ext}^i_R(M,N)$ are its right derived functors, computed in the category of graded modules. They are graded vector spaces, and $\operatorname{Ext}^i_R(M,N(j)) = \operatorname{Ext}^i_R(M,N)(j)$. All homological dimensions are computed in the category of graded modules. We write pd, injdim and gldim for the projective, injective and global dimensions, and GKdim for the Gelfand–Kirillov dimension [33, Chapter 8].

Definition 2.1. A connected graded algebra R is *AS Gorenstein of dimension d* if $\operatorname{injdim} R_R = \operatorname{injdim}{}_R R = d < \infty$ and there is $\ell \in \mathbb{Z}$ such that

$$\operatorname{Ext}^i_R(\Bbbk, R) \cong \begin{cases} \Bbbk(\ell) & i = d, \\ 0 & i \neq d \end{cases},$$

both for right and for left modules. The integers $d = \dim R$ and $\ell = \ell_R$ are the *dimension* and the *Gorenstein parameter* of R. If moreover $\operatorname{gldim} R = d$ and $\operatorname{GKdim} R < \infty$, then R is *AS regular of dimension d*.

The polynomial ring $\Bbbk[x_1,\dots,x_n]$ with $\deg x_i = 1$ is AS regular of dimension n with Gorenstein parameter n. For $q \in \Bbbk^\times$ the quantum plane $\Bbbk_q[x,y] = \Bbbk\langle x,y\rangle/(yx - qxy)$ is AS regular of dimension two with Gorenstein parameter two.

We use the following standard facts about a right noetherian connected graded algebra R. Every finitely generated graded right R-module M has a minimal graded free resolution $P_\bullet \to M$ by finitely generated free modules, obtained from the graded Nakayama lemma, and minimality means that the differentials of $P_\bullet \otimes_R \Bbbk$ vanish. Hence $\operatorname{Tor}^R_i(M,\Bbbk) \cong P_i \otimes_R \Bbbk$, and $\operatorname{pd} M = \max\{i \mid \operatorname{Tor}^R_i(M,\Bbbk) \neq 0\} \leq \operatorname{pd}{}_R\Bbbk$.

The graded Baer criterion shows that $\operatorname{gldim} R$ is the supremum of $\operatorname{pd} R/I$ over the graded right ideals I, so $\operatorname{gldim} R = \operatorname{pd} \Bbbk_R = \operatorname{pd}{}_R\Bbbk$. The same criterion gives

$$\operatorname{injdim} R_R \leq n \iff \operatorname{Ext}^{n+1}_R(R/I, R) = 0 \text{ for every graded right ideal } I. \tag{2.2}$$

If $p = \operatorname{pd}\Bbbk_R < \infty$ and $P_\bullet$ is a minimal resolution of $\Bbbk_R$, then the image of $\operatorname{Hom}_R(P_{p-1}, R) \to \operatorname{Hom}_R(P_p, R)$ lies in $R_+ \operatorname{Hom}_R(P_p, R)$, so the graded Nakayama lemma gives

$$\operatorname{Ext}^p_R(\Bbbk, R) \neq 0 \quad \text{for } p = \operatorname{pd}\Bbbk_R < \infty. \tag{2.3}$$

Let R be a $\Bbbk$-algebra and σ a $\Bbbk$-algebra endomorphism of R. A *σ-derivation* of R is a $\Bbbk$-linear map $\delta\colon R \to R$ such that

$$\delta(ab) = \sigma(a)\delta(b) + \delta(a)b \quad \text{for all } a,b \in R. \tag{2.4}$$

For $u \in R$ the map $\delta_u(a) = ua - \sigma(a)u$ is a σ-derivation, called *inner*, and a σ-derivation is *outer* if it is not inner. The *Ore extension* $R[t;\sigma,\delta]$ is the $\Bbbk$-algebra which is a free left R-module with basis $\{t^n\}_{n\geq 0}$, with multiplication determined by

$$ta = \sigma(a)t + \delta(a) \quad \text{for all } a \in R, \tag{2.5}$$

and we write $R[t;\sigma]$ when $\delta = 0$ [33, §1.2]. For $0 \neq f = \sum_i a_i t^i$ with $a_i \in R$, the *t-degree* $\deg_{\mathrm{t}} f$ is the largest n with $a_n \neq 0$, and $\mathrm{lc}(f) = a_n$ is the *leading coefficient*. An induction on n using (2.5) gives, for $a \in R$ and $n \geq 1$,

$$t^n a \in \sigma^n(a) t^n + \sum_{0<j<n} R t^j + \delta^n(a). \tag{2.6}$$

Multiplying (2.6) on the left by t gives the terms $\sigma^{n+1}(a)t^{n+1} + \delta\sigma^n(a)t^n$, terms in $\sum_{0<j<n+1} Rt^j$, and $t\delta^n(a) = \sigma\delta^n(a)t + \delta^{n+1}(a)$. By (2.6), the coefficient of t^{n+k} in $at^n \cdot bt^k$ is $a\sigma^n(b)$, and all other terms have lower t-degree. So if R is a domain and σ is injective, then

$$\deg_{\mathrm{t}}(fg) = \deg_{\mathrm{t}} f + \deg_{\mathrm{t}} g, \quad \mathrm{lc}(fg) = \mathrm{lc}(f)\, \sigma^{\deg_{\mathrm{t}} f}\big(\mathrm{lc}(g)\big), \tag{2.7}$$

for nonzero f, g, and $R[t;\sigma,\delta]$ is a domain.

Let now R be connected graded, σ a graded automorphism and δ a σ-derivation satisfying (1.4) for some $m \geq 1$. The *graded Ore extension* $S = R[t;\sigma,\delta]$ is graded by $\deg t = m$, that is,

$$S_n = \bigoplus_{k \geq 0} R_{n-mk}\, t^k. \tag{2.8}$$

For $a \in R_n$, both sides of (2.5) have degree $n+m$, so (2.8) makes S a connected graded algebra, with Hilbert series

$$H_S(s) = \frac{H_R(s)}{1 - s^m}. \tag{2.9}$$

If R is right noetherian, then so is S by the skew Hilbert basis theorem [33, Theorem 1.2.9]. Conversely, if S is right noetherian then so is R, since $JS = \bigoplus_n Jt^n$ and hence $JS \cap R = J$ for every right ideal J of R.

Graded Ore extensions preserve AS regularity. A connected graded algebra is AS regular if and only if it is twisted Calabi–Yau in the graded sense [39, Lemma 1.2], and twisted Calabi–Yau algebras are preserved by Ore extensions, with the dimension raised by one, by the graded version of [30, Theorem 3.3], whose proof applies verbatim by [30, Remark 2.2]. Since S is generated by t and generators of R, (2.9) gives $\mathrm{GKdim}\, S \leq \mathrm{GKdim}\, R + 1$. Hence, if R is noetherian and AS regular, then

$$S = R[t;\sigma,\delta] \text{ is AS regular and } \dim S = \dim R + 1. \tag{2.10}$$

An *iterated Ore extension* is an algebra of the form (1.5). We put $B^{(0)} = \Bbbk$, $B^{(1)} = \Bbbk[x_1]$, and for $2 \leq i \leq n$ we require that σ_i be a graded automorphism of

$$B^{(i-1)} = \Bbbk[x_1][x_2;\sigma_2,\delta_2]\cdots[x_{i-1};\sigma_{i-1},\delta_{i-1}],$$

and that δ_i be a σ_i-derivation of $B^{(i-1)}$ raising degrees by one. It is convenient to regard $B^{(1)}$ as the Ore extension $\Bbbk[x_1;\sigma_1,\delta_1]$ with $\sigma_1 = \mathrm{id}_{\Bbbk}$ and $\delta_1 = 0$. By (2.10) and (2.7), every $B^{(i)}$ is a noetherian AS regular domain of dimension i generated in degree one, with Gorenstein parameter i by Corollary 3.11, and with basis $x_1, \ldots, x_i$ of $(B^{(i)})_1$. The monomials $x_1^{a_1}\cdots x_i^{a_i}$ form a basis of $B^{(i)}$, and since $\sigma_i(x_j)$ and $\delta_i(x_j)$ are expressions in $x_1, \ldots, x_{i-1}$, the algebra B is presented by the generators $x_1, \ldots, x_n$ and the relations

$$x_i x_j = \sigma_i(x_j)x_i + \delta_i(x_j) \quad (1 \leq j < i \leq n). \tag{2.11}$$

Example 2.12. The quantum plane is $\Bbbk[x][y;\sigma]$ with $\sigma(x) = qx$. The *Jordan plane* $J = \Bbbk\langle x, y\rangle/(yx - xy - x^2)$ is $\Bbbk[x][y;\delta]$ with $\delta(x) = x^2$. Since $\delta(x^n) = nx^{n+1}$, the relation (2.5) for $a = x^n$ reads

$$yx^n = x^n y + nx^{n+1} \quad (n \geq 0).$$

For $n = 1$ this gives $yx = x(x+y)$ and $xy = (y-x)x$, so $xJ = Jx$ and x is normal. On the other hand $yx = xy + x^2$ does not lie in $Jy \cap J_2 = \Bbbk xy + \Bbbk y^2$, so y is not normal. Both planes have Hilbert series $(1-s)^{-2}$, dimension two and Gorenstein parameter two.

Let G be a group. A *G-grading* of a $\Bbbk$-algebra R is a decomposition $R = \bigoplus_{g \in G} R_g$ into subspaces with $R_g R_h \subseteq R_{gh}$, and $\mathrm{Supp}(R) = \{g \in G \mid R_g \neq 0\}$ is its support. If R is $\mathbb{N}$-graded, the G-grading *refines* the $\mathbb{N}$-grading if $R_n = \bigoplus_g (R_n \cap R_g)$ for all n, and it is *faithful* if $\mathrm{Supp}(R)$ generates G. If R is a domain, then $\mathrm{Supp}(R)$ is closed under products, and a submonoid of a finite group is a subgroup. So a faithful grading of a domain by a finite group has support equal to the group. If the grading refines the $\mathbb{N}$-grading, then $R_0 = \Bbbk \subseteq R_e$.

The identity component of a G-graded right noetherian ring is right noetherian. Indeed, for a right ideal J of R_e, the right ideal $JR = \bigoplus_g JR_g$ of R satisfies $(JR)_e = J$, so $J \mapsto JR$ preserves strict inclusions and ascending chains of right ideals of R_e stabilize.

Let R be G-graded and $\gamma \in G$. A pair (σ, δ) formed by an endomorphism σ of R and a σ-derivation δ is *γ-compatible* if

$$\sigma(R_g) \subseteq R_{\gamma g \gamma^{-1}} \quad \text{and} \quad \delta(R_g) \subseteq R_{\gamma g} \quad \text{for all } g \in G. \tag{2.13}$$

By [2, Theorem A], the grading of R extends to a grading of $R[t; \sigma, \delta]$ in which t is homogeneous of degree γ if and only if (σ, δ) is γ-compatible, and then the extension is unique and given by

$$R[t; \sigma, \delta]_\beta = \bigoplus_{n \geq 0} R_{\beta \gamma^{-n}}\, t^n \quad \text{for all } \beta \in G.$$

Recall that a grading of $A[t; \sigma, \delta]$ is compatible with the Ore structure if A is a graded subalgebra and t is homogeneous. For an iterated Ore extension (1.5), every grading in which $x_1, \dots, x_n$ are homogeneous makes every $B^{(i)}$ a graded subalgebra, since $B^{(i)}$ is generated by $x_1, \dots, x_i$. Such a grading is therefore compatible with each Ore extension $B^{(i)} = B^{(i-1)}[x_i; \sigma_i, \delta_i]$, and it refines the $\mathbb{N}$-grading since the x_i have degree one.

Let G be finite. A G-grading of R is the same as a right $\Bbbk G$-comodule algebra structure $\rho\colon R \to R \otimes \Bbbk G$, given by $\rho(a) = \sum_g a_g \otimes g$ for $a = \sum_g a_g$ with $a_g \in R_g$, and the coinvariants of ρ form R_e. Dually, it is the same as a left module algebra structure over $\Bbbk^G = \mathrm{Hom}_\Bbbk(\Bbbk G, \Bbbk)$, in which the idempotent p_g dual to g acts as the projection onto R_g. If G is abelian and $\Bbbk$ is algebraically closed of characteristic zero, then $\Bbbk^G \cong \Bbbk G^\vee$ for the dual group $G^\vee = \mathrm{Hom}(G, \Bbbk^\times)$, and a G-grading is an action of $G^\vee$ by automorphisms, in which $\chi \in G^\vee$ acts on R_g by $\chi(g)$. Then $R_e = R^{G^\vee}$. This explains the terminology of the introduction.

If $B = \Bbbk[V]$ and G is abelian, then G is a dual reflection group for B if and only if the image of $G^\vee$ in $GL(V)$ is generated by pseudo-reflections [41, 12]. We recall the structure theory of Kirkman, Kuzmanovich and Zhang. Let G be a dual reflection group for B. For every $g \in G$ there is an $\mathbb{N}$-homogeneous element $f_g \in B_g$, with $f_e = 1$, such that

$$B_g = f_g B_e = B_e f_g, \tag{2.14}$$

and the lowest nonzero homogeneous component of B_g is the line $\Bbbk f_g$ [27, Theorems 0.3(1) and 3.5(1)]. The mass element $m_G \in G$ is the unique element with

$$\deg f_{m_G} > \deg f_g \quad \text{for all } g \neq m_G \tag{2.15}$$

[27, Theorem 0.3(4)]. We put

$$p_G(s) = \sum_{g \in G} s^{\deg f_g}, \quad N_G = \deg f_{m_G}.$$

Recall that the covariant ring of G is

$$C(B) = B/B(B_e)_{\geq 1}, \quad (B_e)_{\geq 1} = \bigoplus_{n \geq 1} (B_e)_n,$$

and we write $\bar{b}$ for the image of $b \in B$ in $C(B)$. Kirkman, Kuzmanovich and Zhang proved that $B(B_e)_{\geq 1}$ is a two-sided ideal, that the images of the elements f_g form a basis of $C(B)$, that $C(B)$ is a Frobenius algebra with Hilbert series $p_G(s)$, which is palindromic, and that f_{m_G} is a normal element of B [27, Theorems 0.4, 0.5(1) and 3.5]. That is,

$$\begin{gathered} B(B_e)_{\geq 1} = (B_e)_{\geq 1} B, \quad C(B) = \bigoplus_{g \in G} \Bbbk \bar{f}_g, \\ H_{C(B)}(s) = p_G(s) = s^{N_G} p_G(s^{-1}), \quad f_{m_G} B = B f_{m_G}. \end{gathered} \tag{2.16}$$

In particular $\dim_\Bbbk C(B) = |G|$, and $C(B)$ is G-graded with one-dimensional components $C(B)_g = \Bbbk \bar{f}_g$. Kirkman and Zhang call f_{m_G} the *Jacobian* of the coaction [23, Definition 2.1], and they proved that for every $g \in G$ there is $\mu_g \in \Bbbk^\times$ with

$$f_g f_{g^{-1} m_G} = \mu_g\, f_{m_G} \tag{2.17}$$

[23, Lemma 2.11(2)], applied to $g^{-1} m_G$. Comparing degrees, $\deg f_g + \deg f_{g^{-1} m_G} = N_G$, which is the palindromicity of p_G.

We record the elementary consequences of (2.14) that we use. Let $g \in G$. If $g \neq e$, then $B_g \cap B_0 = 0$ since $B_0 \subseteq B_e$, so $\deg f_g \geq 1$. Since B is a domain, left multiplication by f_g is injective, and

$$B_n \cap B_g = f_g\,(B_e)_{n-\deg f_g} \quad (n \geq 0). \tag{2.18}$$

Summing the Hilbert series of the components gives

$$H_B(s) = p_G(s)\,H_{B_e}(s). \tag{2.19}$$

If $g \neq e$, then (2.18) shows that $B_1 \cap B_g$ is $\Bbbk f_g$ if $\deg f_g = 1$ and zero otherwise. If moreover $(B_e)_1 = 0$ and $B_1 \cap B_g \neq 0$, then $B_2 \cap B_g = f_g(B_e)_1 = 0$. We record this as

$$\dim(B_1 \cap B_g) \leq 1 \quad (g \neq e), \quad (B_e)_1 = 0 \text{ and } B_1 \cap B_g \neq 0 \implies B_2 \cap B_g = 0. \tag{2.20}$$

The first statement is also [27, Theorems 0.3(2) and 3.5(3)]. If θ is a graded automorphism of B with $\theta(B_g) = B_g$, then $\theta(f_g) = \lambda_g f_g$ for some $\lambda_g \in \Bbbk^\times$, since $\Bbbk f_g$ is the lowest component of B_g.

Example 2.21. Let $B = \Bbbk[x_1, \ldots, x_n]$ and $G = \prod_{i=1}^n \langle c_i \rangle$ with $|c_i| = r_i$, and let x_i have G-degree c_i. Then

$$B_e = \Bbbk[x_1^{r_1}, \ldots, x_n^{r_n}], \quad f_g = x_1^{a_1} \cdots x_n^{a_n} \quad \text{for } g = c_1^{a_1} \cdots c_n^{a_n},\ 0 \leq a_i < r_i,$$

so G is a dual reflection group, $m_G = c_1^{r_1-1} \cdots c_n^{r_n-1}$ and $f_{m_G} = \prod_i x_i^{r_i-1}$. Since $(1-s)[r]_s = 1 - s^r$, the identity (2.19) reads

$$\frac{1}{(1-s)^n} = \prod_{i=1}^n [r_i]_s \cdot \prod_{i=1}^n \frac{1}{1-s^{r_i}}, \quad p_G(s) = \prod_{i=1}^n [r_i]_s.$$

The Jacobian of the basic invariants is $\det\left(\partial x_j^{r_j}/\partial x_i\right) = \prod_i r_i x_i^{r_i-1}$, a nonzero multiple of f_{m_G}. Here G is the character group of the reflection group $\prod_i \mu_{r_i}$ acting diagonally on $\Bbbk^n$, and $C(B) = \bigotimes_i \Bbbk[x_i]/(x_i^{r_i})$.

Example 2.22. Let $A = \Bbbk_q[x, y]$ be the quantum plane, graded by $\mathbb{Z}_2 = \{e, \tau\}$ with $x \in A_\tau$ and $y \in A_e$. Then A_e is spanned by the monomials $x^{2a}y^b$, and $yx^2 = q^2x^2y$, so

$$A_e = \Bbbk\langle x^2, y\rangle \cong \Bbbk_{q^2}[x^2, y], \quad f_\tau = x, \quad m_G = \tau, \quad C(A) = \Bbbk[\bar{x}]/(\bar{x}^2).$$

So $\mathbb{Z}_2$ is a dual reflection group for A, and

$$H_A(s) = \frac{1}{(1-s)^2} = (1+s) \cdot \frac{1}{(1-s^2)(1-s)} = p_G(s)\,H_{A_e}(s).$$

The ideal $A(A_e)_{\geq 1}$ is generated by x^2 and y. The nontrivial character of $\mathbb{Z}_2$ acts by $x \mapsto -x$ and $y \mapsto y$, and for $q = 1$ this is the reflection of $\Bbbk^2$ in the line $x = 0$.

3. Descent of the Gorenstein property

In this section we prove Theorem A. Throughout, R is a noetherian connected graded algebra, σ is a graded automorphism of R, δ is a σ-derivation satisfying (1.4) for a fixed $m \geq 1$, and $S = R[t; \sigma, \delta]$ is graded by $\deg t = m$. By Section 2, S is a right noetherian connected graded algebra, and it is left noetherian by the left version of the same argument, since σ is an automorphism [33, Theorem 1.2.9]. The proof compares $\mathrm{Ext}_R(\Bbbk, R)$ with Ext_S of the cyclic module S/R_+S.

Lemma 3.1. *The following hold.*

(1) *The algebra S is free as a left and as a right R-module on the powers of t,*

$$S = \bigoplus_{n\geq 0} Rt^n = \bigoplus_{n\geq 0} t^n R, \quad S_R \cong \bigoplus_{n\geq 0} R(-mn) \cong {}_RS. \tag{3.2}$$

(2) *The ideal $I = R_+S$ is two-sided and equal to SR_+, and $Q = S/I$ is the polynomial ring $\Bbbk[\bar{t}]$ with $\deg \bar{t} = m$. As a right and as a left R-module,*

$$Q \cong \bigoplus_{n\geq 0} \Bbbk(-mn). \tag{3.3}$$

(3) *There are exact sequences of graded right and left S-modules, respectively,*

$$0 \to Q(-m) \xrightarrow{\bar{t}\,\cdot} Q \to \Bbbk_S \to 0 \quad \textit{and} \quad 0 \to Q(-m) \xrightarrow{\cdot\,\bar{t}} Q \to {}_S\Bbbk \to 0. \tag{3.4}$$

Proof. (1) The first decomposition is the definition of S. By (2.6), $t^n a - \sigma^n(a)t^n \in \sum_{j<n} Rt^j$ for $a \in R$. Since σ^n is surjective, induction on n gives $\sum_{j\le n} t^j R = \sum_{j\le n} Rt^j$. If $\sum_{j\le n} t^j a_j = 0$ with $a_n \neq 0$, then the coefficient of t^n in the basis $\{t^j\}$ is $\sigma^n(a_n) \neq 0$, since σ^n is injective, a contradiction. So the sum $\sum_n t^n R$ is direct, and $a \mapsto t^n a$ is an isomorphism of graded right R-modules $R(-mn) \to t^n R$.

(2) Since σ is a graded bijection and δ raises degrees, $\sigma(R_+) = R_+$ and $\delta(R_+) \subseteq R_+$. For $a \in R_+$ put $a' = \sigma^{-1}(a) \in R_+$. Then

$$ta = \sigma(a)t + \delta(a) \in R_+S, \quad at = ta' - \delta(a') \in SR_+.$$

So $tR_+ \subseteq R_+S$ and $R_+t \subseteq SR_+$. Since S is generated by R and t, the right ideal R_+S and the left ideal SR_+ are two-sided, and both equal the ideal generated by R_+. By (1), $R_+S = \bigoplus_n R_+t^n$, so Q has basis $\{\bar{t}^n\}$, with $\deg \bar{t}^n = mn$, and $Q = \Bbbk[\bar{t}]$. Since $QR_+ = 0 = R_+Q$, the algebra R acts on Q through $R/R_+ = \Bbbk$ on both sides, which gives (3.3).

(3) Left multiplication by $\bar{t}$ is injective on $\Bbbk[\bar{t}]$, it raises degrees by m, and it is right S-linear because I is two-sided. Its cokernel is $S/(I + tS) = S/S_+ = \Bbbk$, since $S_+ = R_+S + tS$. The left sequence is obtained in the same way from $S_+ = SR_+ + St$. □

Lemma 3.5. *Let M be a finitely generated graded right R-module. For every $i \ge 0$ there is an isomorphism of graded vector spaces*

$$\operatorname{Ext}^i_S(M \otimes_R S, S) \cong \bigoplus_{n\ge 0} \operatorname{Ext}^i_R(M, R)(-mn).$$

In particular, if $X^i = \operatorname{Ext}^i_R(\Bbbk_R, R)$, then $\operatorname{Ext}^i_S(Q, S) \cong \bigoplus_{n\ge 0} X^i(-mn)$. The same holds for left modules, with $S \otimes_R M$ in place of $M \otimes_R S$.

Proof. Let $P_\bullet \to M$ be a resolution by finitely generated graded free right R-modules. Since ${}_RS$ is free, $P_\bullet \otimes_R S$ is a free resolution of $M \otimes_R S$. For a free right R-module P we have $\operatorname{Hom}_S(P \otimes_R S, S) \cong \operatorname{Hom}_R(P, S_R)$, hence $\operatorname{Ext}^i_S(M \otimes_R S, S) \cong \operatorname{Ext}^i_R(M, S_R)$. By (3.2), $S_R \cong \bigoplus_n R(-mn)$, and $\operatorname{Ext}^i_R(M, -)$ commutes with this direct sum because the P_i are finitely generated. Altogether,

$$\operatorname{Ext}^i_S(M \otimes_R S, S) \cong \operatorname{Ext}^i_R(M, S_R) \cong \operatorname{Ext}^i_R\Big(M, \bigoplus_{n\ge 0} R(-mn)\Big) \cong \bigoplus_{n\ge 0} \operatorname{Ext}^i_R(M, R)(-mn).$$

The second statement is the case $M = \Bbbk$, since $\Bbbk \otimes_R S = S/R_+S = Q$. For left modules, the same argument applies to $S \otimes_R -$, using that S_R is free, so that $S \otimes_R P_\bullet$ is a free resolution of $S \otimes_R M$, and that ${}_RS \cong \bigoplus_n R(-mn)$. Both uses rely on the two free decompositions (3.2), hence on the bijectivity of σ. □

We bound the injective dimension of R with local duality for S. Let T be a noetherian connected graded algebra and N a graded right T-module. The *torsion submodule* of N is

$$\Gamma_{\mathfrak{m}}(N) = \{b \in N \mid bT_{\ge k} = 0 \text{ for some } k \ge 0\},$$

and the *local cohomology* of N is $H^i_{\mathfrak{m}}(N) = \mathrm{R}^i\Gamma_{\mathfrak{m}}(N)$, where $\mathfrak{m} = T_+$, computed in the category of graded modules. The *graded dual* of a graded vector space N is N' with $(N')_n = \operatorname{Hom}_{\Bbbk}(N_{-n}, \Bbbk)$. If N is finitely generated, then $\Gamma_{\mathfrak{m}}(N)$ is a finitely generated submodule annihilated by some $T_{\ge k}$, hence finite-dimensional. If a noetherian connected graded algebra T is AS Gorenstein of dimension D with Gorenstein parameter ℓ, then it has a balanced dualizing complex of the form ${}^{\nu}T(-\ell)[D]$ for a graded automorphism ν of T [47, Section 4], see also [29, Theorem 4.12] and [46]. Then T satisfies the χ-condition and has finite local cohomological dimension [47, Corollary 4.3], see also [29, Corollary 4.13] and [45, Theorem 6.3], and local duality [45, Theorem 5.1] gives isomorphisms of graded vector spaces

$$\operatorname{Ext}^i_T(N, T) \cong H^{D-i}_{\mathfrak{m}}(N)'(\ell), \tag{3.6}$$

for every finitely generated graded right or left T-module N. For $i = D$ this gives $\operatorname{Ext}^D_T(N, T) \cong \Gamma_{\mathfrak{m}}(N)'(\ell)$, which is finite-dimensional. Only this finite-dimensionality is used below.

Proof of Theorem A. If $b \in S$ satisfies $bS_+ = 0$, then $bt = 0$, hence $b = 0$ by (3.2). Thus $\operatorname{Hom}_S(\Bbbk, S) = 0$, and $D \ge 1$.

We compute the spaces $X^i = \operatorname{Ext}^i_R(\Bbbk, R)$ of Lemma 3.5. Since R is noetherian, each X^i is bounded below and finite-dimensional in each degree. Put $Y^i = \operatorname{Ext}^i_S(Q, S)$. Then $\operatorname{Ext}^i_S(Q(-m), S) = Y^i(m)$, and the first sequence of (3.4) gives a long exact sequence of graded vector spaces

$$\cdots \to \operatorname{Ext}^i_S(\Bbbk, S) \to Y^i \xrightarrow{\varphi_i} Y^i(m) \to \operatorname{Ext}^{i+1}_S(\Bbbk, S) \to Y^{i+1} \to \cdots, \tag{3.7}$$

where each φ_i preserves degrees. Since S is AS Gorenstein, $\operatorname{Ext}^j_S(\Bbbk, S) = 0$ for $j \neq D$ and $\operatorname{Ext}^D_S(\Bbbk, S) = \Bbbk(\ell)$.

Let $i \neq D-1$. Then $\operatorname{Ext}^{i+1}_S(\Bbbk, S) = 0$, so φ_i is surjective. Suppose that $X^i \neq 0$, and let a be its lowest degree. By Lemma 3.5, $Y^i \cong \bigoplus_{n\geq 0} X^i(-mn)$ has lowest degree a, so it vanishes in degree $a-m$, while $Y^i(m)$ does not. This contradicts the surjectivity of φ_i, so $X^i = 0$. For $i = D-1$ we have $Y^D = 0$ by the previous case, and (3.7) reduces to the exact sequence

$$0 \to Y^{D-1} \xrightarrow{\varphi_{D-1}} Y^{D-1}(m) \to \Bbbk(\ell) \to 0. \tag{3.8}$$

Let $h(s)$ be the Hilbert series of X^{D-1}. By Lemma 3.5, the Hilbert series of Y^{D-1} and of $Y^{D-1}(m)$ are $h(s)/(1-s^m)$ and $s^{-m}h(s)/(1-s^m)$. Exactness of (3.8) gives

$$\begin{aligned} \frac{s^{-m}h(s)}{1-s^m} - \frac{h(s)}{1-s^m} &= s^{-\ell}, \\ s^{-m}h(s) = s^{-\ell}, \quad h(s) &= s^{m-\ell}. \end{aligned} \tag{3.9}$$

So X^{D-1} is one-dimensional, concentrated in degree $m-\ell$. A one-dimensional graded module is annihilated by R_+, hence $X^{D-1} \cong \Bbbk(\ell - m)$. The second sequence of (3.4) and the left version of Lemma 3.5 give the same statement for left modules. Altogether,

$$\operatorname{Ext}^i_R(\Bbbk, R) \cong \begin{cases} \Bbbk(\ell - m) & i = D-1, \\ 0 & i \neq D-1 \end{cases}, \tag{3.10}$$

on both sides.

It remains to show that $\operatorname{injdim} R_R = \operatorname{injdim}{}_R R = D-1$. Let M be a finitely generated graded right R-module, and put $N = M \otimes_R S$, a finitely generated graded right S-module. By (3.6) with $T = S$, the space $\operatorname{Ext}^D_S(N, S)$ is finite-dimensional. By Lemma 3.5,

$$\operatorname{Ext}^D_S(N, S) \cong \bigoplus_{n\geq 0} \operatorname{Ext}^D_R(M, R)(-mn).$$

If $\operatorname{Ext}^D_R(M, R)$ were nonzero in some degree a, this space would be nonzero in every degree $a + mn$ with $n \geq 0$. Hence $\operatorname{Ext}^D_R(M, R) = 0$ for every such M, so $\operatorname{injdim} R_R \leq D-1$ by (2.2), and equality holds since $X^{D-1} \neq 0$. The left statement is the same, using the left version of Lemma 3.5 and local duality for left modules. This proves that R is AS Gorenstein of dimension $D-1$ with Gorenstein parameter $\ell - m$.

Suppose now that S is AS regular. It remains to show that $\operatorname{gldim} R = D-1$ and $\operatorname{GKdim} R < \infty$. Since S_R is free, every projective right S-module is projective over R. A projective resolution of Q over S of length at most D therefore restricts to a projective resolution of Q_R over R. By (3.3), $\Bbbk_R$ is a direct summand of Q_R, so $p = \operatorname{pd} \Bbbk_R \leq D$. By (2.3), $\operatorname{Ext}^p_R(\Bbbk, R) \neq 0$, so $p = D-1$ by (3.10), and $\operatorname{gldim} R = \operatorname{pd} \Bbbk_R = D-1$. Since R is a subalgebra of S, we have $\operatorname{GKdim} R \leq \operatorname{GKdim} S < \infty$. □

Corollary 3.11. *Let $S = R[t; \sigma, \delta]$ be a graded Ore extension as above. Then S is AS regular if and only if R is AS regular. In this case*

$$\dim S = \dim R + 1 \quad \textit{and} \quad \ell_S = \ell_R + m. \tag{3.12}$$

Proof. The implication from R to S is (2.10), and the converse together with (3.12) is Theorem A. □

Iterating Theorem A gives the descent along iterated Ore extensions, with arbitrary degrees of the variables.

Corollary 3.13. *Let R be a noetherian connected graded algebra, put $S^{(0)} = R$, and for $1 \leq j \leq k$ let*

$$S^{(j)} = S^{(j-1)}[t_j; \sigma_j, \delta_j], \quad \deg t_j = m_j,$$

where σ_j is a graded automorphism of $S^{(j-1)}$ and δ_j is a σ_j-derivation of $S^{(j-1)}$ raising degrees by $m_j \geq 1$. If $S^{(k)}$ is AS Gorenstein of dimension D with Gorenstein parameter ℓ, then R is AS Gorenstein of dimension $D-k$ with Gorenstein parameter

$$\ell_R = \ell - (m_1 + m_2 + \cdots + m_k).$$

If moreover $S^{(k)}$ is AS regular, then R is AS regular.

Proof. Each $S^{(j)}$ is noetherian and connected graded. Applying Theorem A to $S^{(j)} = S^{(j-1)}[t_j; \sigma_j, \delta_j]$ for $j = k, k-1, \dots, 1$ shows that $S^{(j-1)}$ is AS Gorenstein of dimension $D-(k-j+1)$ with Gorenstein parameter $\ell - (m_j + \cdots + m_k)$, and AS regular if $S^{(k)}$ is. □

Remark 3.14. When $\delta = 0$, the element t is normal and regular of degree m in S, and $R \cong S/tS$. The Rees lemma [28, Proposition 3.4(b)] then gives $\operatorname{Ext}^{i+1}_S(\Bbbk, S) \cong \operatorname{Ext}^i_R(\Bbbk, R)(m)$, which is (3.10) directly. The proof above uses no normal element. Its point is the case $\delta \neq 0$, where t is in general not normal. The hypotheses enter as follows. The decomposition $S = \bigoplus_n Rt^n$ holds by definition, while $S = \bigoplus_n t^n R$ uses the bijectivity of σ. Together they give the right and left versions of Lemma 3.5. The noetherian hypothesis gives finitely generated free resolutions, and it makes S noetherian, which is needed for local duality. In the regular case local duality can be avoided, since $\operatorname{gldim} R = D - 1$ follows from (3.10) and (2.3) alone and bounds the injective dimension. The condition $m \geq 1$ makes S connected graded, and the shift by m in (3.4) produces the Gorenstein parameter $\ell - m$.

Example 3.15. Let $S = \Bbbk[x][y; \delta]$ be the Jordan plane of Example 2.12. Here $R = \Bbbk[x]$, $m = 1$, $D = 2$ and $\ell = 2$, and $Q = S/xS = \Bbbk[\bar{y}]$. Since $X^1 = \operatorname{Ext}^1_{\Bbbk[x]}(\Bbbk, \Bbbk[x]) = \Bbbk(1)$ and $X^i = 0$ for $i \neq 1$, Lemma 3.5 gives $Y^1 = \operatorname{Ext}^1_S(Q, S) \cong \bigoplus_{n \geq 0} \Bbbk(1-n)$, with Hilbert series $s^{-1}/(1-s)$. The exact sequence (3.8) reads

$$0 \to Y^1 \xrightarrow{\varphi_1} Y^1(1) \to \Bbbk(2) \to 0, \quad \frac{s^{-2}}{1-s} - \frac{s^{-1}}{1-s} = s^{-2},$$

which is (3.9) with $h(s) = s^{-1} = s^{m-\ell}$. Here R is not the quotient of S by the Ore variable y, which is not normal, so this computation does not come from the Rees lemma.

Example 3.16. The degree of t may exceed one. Let $R = \Bbbk[x]$ and let δ be the derivation with $\delta(x) = x^3$, so that (1.4) holds with $m = 2$. The algebra

$$S = \Bbbk[x][t; \delta] = \Bbbk\langle x, t\rangle/(tx - xt - x^3), \quad \deg x = 1, \quad \deg t = 2,$$

is AS regular of dimension two by (2.10), with Hilbert series $H_S(s) = (1-s)^{-1}(1-s^2)^{-1}$, and (3.12) gives $\ell_S = \ell_{\Bbbk[x]} + 2 = 3$. Here t is not normal, since $tx = xt + x^3$ does not lie in $St \cap S_3 = \Bbbk xt$.

Example 3.17. Theorem A also applies to algebras that are not AS regular. Let $R = \Bbbk[x]/(x^2)$ with $\deg x = 1$, and $S = R[t] = \Bbbk[x,t]/(x^2)$ with $\deg t = 1$. The algebra S is a hypersurface in $\Bbbk[x,t]$, hence a commutative graded Gorenstein ring of dimension one, and its a-invariant is $\deg x^2 - 2 = 0$ [8, §3.6]. In our convention the Gorenstein parameter is $\ell = -a$. So S is AS Gorenstein of dimension one with Gorenstein parameter 0. By Theorem A, R is AS Gorenstein of dimension 0 with Gorenstein parameter -1. This agrees with the direct computation. The algebra R is self-injective, and $\operatorname{Hom}_R(\Bbbk, R)$ is spanned by the inclusion of the socle $\Bbbk x$, which lies in degree one, so $\operatorname{Hom}_R(\Bbbk, R) \cong \Bbbk(-1)$.

4. Dual reflection groups of an Ore extension

In this section we prove Theorem B, describe the elements f_g of B, and discuss the compatibility assumption. From now on $\Bbbk$ is algebraically closed of characteristic zero.

Throughout this section, A is a noetherian AS regular domain generated in degree one, σ is a graded automorphism of A, δ is a σ-derivation with $\delta(A_n) \subseteq A_{n+1}$, and $B = A[t; \sigma, \delta]$ with $\deg t = 1$. By (2.10) and (2.7), B is a noetherian AS regular domain generated in degree one, and the t-degree is additive on B. We fix a finite group G and a G-grading of B compatible with the Ore structure, with $t \in B_\gamma$, and we put

$$K = \operatorname{Supp}(A), \quad r = |\gamma|.$$

Since A is a domain, K is a subgroup of G. The grading of A refines its $\mathbb{N}$-grading, since A is a graded subalgebra of B. By Section 2, the pair (σ, δ) is γ-compatible and

$$B_\beta = \bigoplus_{n \geq 0} A_{\beta\gamma^{-n}}\, t^n \quad \text{for all } \beta \in G. \tag{4.1}$$

Lemma 4.2. *The following hold for every grading of B compatible with the Ore structure.*

(1) *$\sigma(A_g) = A_{\gamma g \gamma^{-1}}$ for every $g \in G$. In particular $\sigma(A_e) = A_e$, and $\gamma K \gamma^{-1} = K$.*
(2) *$\operatorname{Supp}(B) = K\langle\gamma\rangle$. Hence $G = K\langle\gamma\rangle$, the subgroup K is normal in G, and $|G| = |K|\, r/|K \cap \langle\gamma\rangle|$.*

(3) *If $\delta \neq 0$, then $\gamma \in K$.*

Proof. (1) The equality $\sigma(A_g) = A_{\gamma g \gamma^{-1}}$ is [2, Lemma 2.3(3)]. Since σ is injective, $A_g \neq 0$ if and only if $A_{\gamma g \gamma^{-1}} \neq 0$.

(2) By (4.1), $B_\beta \neq 0$ if and only if $\beta\gamma^{-n} \in K$ for some $n \geq 0$, that is, $\beta \in K\gamma^n$. Since γ has finite order, $\mathrm{Supp}(B) = \bigcup_n K\gamma^n = K\langle\gamma\rangle$. By (1), γ normalizes K, so $K\langle\gamma\rangle$ is a subgroup in which K is normal, and faithfulness gives $G = K\langle\gamma\rangle$. The order formula is the product formula for the subgroups K and $\langle\gamma\rangle$.

(3) Since $\delta \neq 0$ and $A = \bigoplus_g A_g$, there is a homogeneous $a \in A_g$ with $\delta(a) \neq 0$. By (2.13), $0 \neq \delta(a) \in A_{\gamma g}$. So $g \in K$ and $\gamma g \in K$, and $\gamma = (\gamma g)g^{-1} \in K$. □

Lemma 4.3. *If G is a dual reflection group for B, then $K \cap \langle\gamma\rangle = \{e\}$.*

Proof. Suppose that $h = \gamma^a \in K$ with $h \neq e$ and $a \geq 1$. Let f_h be as in (2.14) for B. Choose $0 \neq b \in A_h$ and write $b = f_h c$ with $c \in B_e$. By (2.7), $\deg_t f_h + \deg_t c = \deg_t b = 0$, so $f_h \in A$, and $\deg f_h \geq 1$ since $h \neq e$. On the other hand $t^a \in B_h = f_h B_e$, so

$$t^a = f_h c' = \sum_i f_h c_i' t^i \quad \text{with } c' = \sum_i c_i' t^i \in B_e,\ c_i' \in A.$$

Comparing the coefficients of t^a in the free left A-module $\bigoplus_i At^i$ gives $f_h c_a' = 1$ in A. This is impossible, since A is connected graded and $f_h \in A_{\geq 1}$. □

The same argument appears in the proof of [11, Proposition 6.1] when $\delta = 0$. Here we only use that the t-degree is additive, which holds for every δ. Suppose now that $K \cap \langle\gamma\rangle = \{e\}$. For $k \in K$ and $0 \leq j < r$, the element $k\gamma^{j-n}$ lies in K if and only if r divides $n - j$. So (4.1) gives

$$B_{k\gamma^j} = \bigoplus_{n \geq 0} A_{k\gamma^{j-n}}\, t^n = \bigoplus_{l \geq 0} A_k\, t^{j+rl} \quad (k \in K,\ 0 \leq j < r). \tag{4.4}$$

In particular $B_e = \bigoplus_{l \geq 0} A_e t^{rl}$. If $\delta \neq 0$, then $\gamma \in K$ by Lemma 4.2(3), so $\gamma = e$ and $r = 1$. In this case $\sigma(A_e) = A_e$ and $\delta(A_e) \subseteq A_\gamma = A_e$. If $\delta = 0$, then $t^r a = \sigma^r(a)t^r$ for $a \in A$. In both cases

$$B_e = A_e[t^r; \sigma^r|_{A_e}, \delta|_{A_e}], \quad \deg t^r = r, \tag{4.5}$$

is a graded Ore extension of A_e, where $\delta = 0$ whenever $r > 1$. The algebra A_e is noetherian by Section 2, since A is.

Proof of Theorem B. Assume that G is a dual reflection group for B. By Lemma 4.3, $K \cap \langle\gamma\rangle = \{e\}$, so (4.5) holds. The algebra B_e is AS regular, A_e is noetherian, $\sigma^r|_{A_e}$ is a graded automorphism of A_e, and $\delta|_{A_e}$ raises degrees by r since it vanishes when $r > 1$. So Theorem A applies with $m = r$ and shows that A_e is AS regular. The grading of A by K is faithful and refines the $\mathbb{N}$-grading. Hence K is a dual reflection group for A.

Conversely, assume that K is a dual reflection group for A and that $K \cap \langle\gamma\rangle = \{e\}$. Then (4.5) exhibits B_e as a graded Ore extension of the AS regular algebra A_e, so B_e is AS regular by (2.10). The grading of B is faithful and refines the $\mathbb{N}$-grading by our conventions, so G is a dual reflection group for B.

In this case K is normal in $G = K\langle\gamma\rangle$ by Lemma 4.2(2), and $K \cap \langle\gamma\rangle = \{e\}$, so $G = K \rtimes \langle\gamma\rangle$. If $\delta \neq 0$, we saw that $\gamma = e$, and then (2.13) shows that σ and δ preserve every component A_g. It remains to prove that $(A_e)_1 \neq 0$ when $\delta \neq 0$. Suppose that $(A_e)_1 = 0$. Then $A_1 = \bigoplus_{g \neq e}(A_1 \cap A_g)$. Let $g \neq e$ with $A_1 \cap A_g \neq 0$. By (2.20) applied to A, we have $A_2 \cap A_g = 0$. Since δ preserves A_g and raises degrees by one,

$$\delta(A_1 \cap A_g) \subseteq A_2 \cap A_g = 0.$$

Hence δ vanishes on A_1. Since A is generated by A_1, (2.4) gives $\delta = 0$, a contradiction. □

We describe the elements f_g of B in terms of those of A.

Proposition 4.6. *In the setting of Theorem B, suppose that G is a dual reflection group for B, and let f_k, for $k \in K$, be the elements of* (2.14) *for A. Then for $k \in K$ and $0 \leq j < r$,*

$$B_{k\gamma^j} = f_k t^j\, B_e = B_e\, f_k t^j. \tag{4.7}$$

So the elements $f_{k\gamma^j} = f_k t^j$ play the role of (2.14) *for B, and*

$$p_G(s) = p_K(s)\,[r]_s, \quad m_G = m_K \gamma^{r-1}, \quad f_{m_G} = f_{m_K} t^{r-1}, \quad N_G = N_K + r - 1. \tag{4.8}$$

Proof. By (4.4) and (2.14) for A,

$$B_{k\gamma^j} = \bigoplus_{l\geq 0} f_k A_e\, t^{j+rl}.$$

If $r = 1$, then $j = 0$. If $r > 1$, then $\delta = 0$ and $t^j A_e = \sigma^j(A_e)t^j = A_e t^j$ by Lemma 4.2(1). In both cases $A_e t^j = t^j A_e$, hence $f_k A_e t^{j+rl} = f_k t^j A_e t^{rl}$, and $B_{k\gamma^j} = f_k t^j B_e$. For the other side, $B_e\, f_k t^j = \bigoplus_l A_e t^{rl} f_k t^j$. If $r > 1$, then $\delta = 0$ and $t^{rl} f_k = \sigma^{rl}(f_k)t^{rl}$, and $\sigma^{rl}(f_k)$ is a nonzero multiple of f_k, since σ^r is a graded automorphism preserving A_k by Lemma 4.2(1) and $\Bbbk f_k$ is the lowest component of A_k. If $r = 1$, then $j = 0$ and $B_e f_k \subseteq B_k = f_k B_e$, and both sides have Hilbert series $s^{\deg f_k} H_{B_e}(s)$ because B is a domain. In both cases $B_e f_k t^j = B_{k\gamma^j}$. The lowest component of $B_{k\gamma^j}$ is therefore $\Bbbk f_k t^j$, of degree $\deg f_k + j$. Since every element of $G = K \rtimes \langle\gamma\rangle$ is uniquely of the form $k\gamma^j$ with $0 \leq j < r$, summing over G gives $p_G(s) = p_K(s)[r]_s$, and the maximal degree is attained exactly at $k = m_K$ and $j = r - 1$. □

When $\delta \neq 0$, the elements f_g of B are those of A, and the σ-derivation acts trivially on the covariant ring of A, as follows.

Proposition 4.9. *In the setting of Theorem B, suppose that G is a dual reflection group for B and that $\delta \neq 0$. For every $g \in G$ there are $\lambda_g \in \Bbbk^\times$ and $a_g \in (A_e)_1$ with*

$$\sigma(f_g) = \lambda_g f_g, \quad \delta(f_g) = f_g a_g, \quad t f_g = f_g(\lambda_g t + a_g). \tag{4.10}$$

Moreover $\delta(A) \subseteq A(A_e)_{\geq 1}$, so δ induces the zero map on $C(A)$.

Proof. By Theorem B, σ and δ preserve A_g. Since σ preserves degrees, $\sigma(f_g) = \lambda_g f_g$ with $\lambda_g \neq 0$. The element $\delta(f_g) \in A_g$ has degree $\deg f_g + 1$, so $\delta(f_g) = f_g a_g$ with $a_g \in (A_e)_1$ by (2.18). Then $t f_g = \sigma(f_g)t + \delta(f_g) = f_g(\lambda_g t + a_g)$. For $b \in A_e$, the element $\delta(b)$ lies in $(A_e)_{\geq 1}$, since δ preserves A_e and raises degrees. Hence

$$\delta(f_g b) = \sigma(f_g)\delta(b) + \delta(f_g)b = f_g\big(\lambda_g \delta(b) + a_g b\big) \in f_g(A_e)_{\geq 1}.$$

Since $A = \bigoplus_g f_g A_e$, this proves $\delta(A) \subseteq A(A_e)_{\geq 1}$. □

The relation $B_e f_g = f_g B_e$ of (4.7) defines a graded automorphism η_g of B_e by $b f_g = f_g \eta_g(b)$, and (4.10) says that $\eta_g(t) = \lambda_g t + a_g$. This is the automorphism η_g of [27, (E3.5.1)].

The first example is one of the two Ore extensions with $\delta \neq 0$ in the list of Buzaglo and Rogalski.

Example 4.11. Let $A_{-1} = \Bbbk\langle u, v\rangle/(u^2 - v^2)$, let σ be the automorphism exchanging u and v, and let δ be the σ-derivation with $\delta(u) = uv$ and $\delta(v) = vu$. Then

$$\delta(u^2 - v^2) = \sigma(u)\delta(u) + \delta(u)u - \sigma(v)\delta(v) - \delta(v)v = vuv + uvu - uvu - vuv = 0,$$

so δ is well defined, and $B = A_{-1}[z; \sigma, \delta]$ is the algebra with relations

$$u^2 = v^2, \quad zu = vz + uv, \quad zv = uz + vu$$

of [11, Table 1]. Consider a grading of B by a finite group G for which u, v, z are homogeneous, of degrees a, b, c. By [11, Theorem 3.4], every grading of B by a nonabelian group is of this form up to a graded change of variables. The subalgebra A_{-1} is graded, so the grading is compatible with the Ore structure, with $\gamma = c$. By (2.13) applied to $\sigma(u) = v$ and $\delta(u) = uv$, we get $b = cac^{-1}$ and $ab = ca$. If G is a dual reflection group for B, then $c = e$ by Theorem B, hence $b = a$ and $a^2 = a$. So $a = b = c = e$ and G is trivial. Hence B has no nonabelian dual reflection group, in agreement with [11, Theorem 1.2], and the argument uses the classification of gradings [11, Theorem 3.4] but not the classification of dual reflection groups.

On the other hand, B has the $\mathbb{Z}_2$-grading of [11, Remark 5.16], defined in the basis $x = (u + v)/2$, $y = (u - v)/2$, z. In this basis $u^2 = v^2$ reads $yx = -xy$, and adding and subtracting the two other relations gives

$$yx = -xy, \quad zx = xz + x^2 - y^2, \quad zy = -yz - 2xy. \tag{4.12}$$

So B is the iterated Ore extension $\Bbbk[x][y; \sigma_2][z; \sigma_3, \delta_3]$ with $\sigma_2(x) = -x$, $\sigma_3(x) = x$, $\sigma_3(y) = -y$, $\delta_3(x) = x^2 - y^2$ and $\delta_3(y) = -2xy$. Write $\mathbb{Z}_2 = \{e, \tau\}$ and grade B by $x, z \in B_e$ and $y \in B_\tau$. The subalgebra $A = \Bbbk\langle x, y\rangle$ is graded, z is homogeneous of degree $\gamma = e$, and σ_3, δ_3 preserve both components. Put $w = y^2$. Since $yx = -xy$, the element w is central in A, and $A_e = \Bbbk[x, w]$ is a polynomial ring with $\deg w = 2$. So $\mathbb{Z}_2$ is a dual reflection group for A with $f_\tau = y$, and by Theorem B also for B, with $B_e = A_e[z; \sigma_3|_{A_e}, \delta_3|_{A_e}]$. We have $\sigma_3(w) = w$, and using $zy = -yz - 2xy$ twice,

$$zw = (zy)y = -y(zy) - 2xyy = -y(-yz - 2xy) - 2xw = wz + 2yxy - 2xw = wz - 4xw.$$

Hence

$$B_e = \Bbbk[x,w][z;\delta_e], \quad \delta_e(x) = x^2 - w, \quad \delta_e(w) = -4xw, \tag{4.13}$$

with $\deg x = \deg z = 1$ and $\deg w = 2$. Its Hilbert series is $(1-s)^{-2}(1-s^2)^{-1}$, and $\ell_{B_e} = 4 = \ell_B + \deg f_\tau$. Here $(A_e)_1 = \Bbbk x \neq 0$, as predicted by Theorem B, and in (4.10) we have $\lambda_\tau = -1$ and $\delta_3(y) = y \cdot (2x)$, so $a_\tau = 2x$ and $\eta_\tau(z) = -z + 2x$.

Example 4.14. Let $\alpha \in \Bbbk^\times$ and $A = \Bbbk_{-1}[x,y] = \Bbbk\langle x,y\rangle/(yx+xy)$. Let σ be the graded automorphism with $\sigma(x) = \alpha x$ and $\sigma(y) = -y$, and let δ be the σ-derivation of the free algebra with

$$\delta(x) = y^2, \quad \delta(y) = 0.$$

By (2.4), $\delta(yx+xy) = \sigma(y)\delta(x) + \delta(y)x + \sigma(x)\delta(y) + \delta(x)y = -y^3 + y^3 = 0$, so δ induces a σ-derivation of A raising degrees by one. This derivation is outer. If $\delta = \delta_u$, then also $\delta = \delta_{u_1}$ for the component u_1 of u of degree one, since δ raises degrees by one. For $u = px + qy \in A_1$,

$$\delta_u(x) = ux - \sigma(x)u = p(1-\alpha)x^2 - q(1+\alpha)xy,$$

which is never y^2. Put $B = A[z;\sigma,\delta]$, that is,

$$B = \Bbbk\langle x,y,z\rangle/\big(yx + xy,\ zx - \alpha xz - y^2,\ zy + yz\big).$$

Grade B by $\mathbb{Z}_2 = \{e,\tau\}$ with $x, z \in B_e$ and $y \in B_\tau$. This grading is compatible with the Ore structure, with $\gamma = e$. Since $A_e = \Bbbk[x,w]$ with $w = y^2$ central in A, the group $\mathbb{Z}_2$ is a dual reflection group for A, and by Theorem B also for B. We have $\sigma(w) = \sigma(y)^2 = w$ and $\delta(w) = \sigma(y)\delta(y) + \delta(y)y = 0$, so

$$B_e = \Bbbk[x,w][z;\sigma|_{A_e},\delta|_{A_e}], \quad zx = \alpha xz + w, \quad zw = wz.$$

The algebra B_e is AS regular of dimension three with generators of degrees $1,2,1$, so $H_{B_e}(s) = (1-s)^{-2}(1-s^2)^{-1}$ and $\ell_{B_e} = 4 = \ell_B + \deg f_\tau$, with $f_\tau = y$, $m_G = \tau$ and $(A_e)_1 = \Bbbk x \neq 0$. In (4.10) we have $\lambda_\tau = -1$ and $a_\tau = 0$, so $\eta_\tau(z) = -z$.

Example 4.15. Let $J = \Bbbk[x][y;\delta]$ be the Jordan plane, with $\delta(x) = x^2$. Every grading of J compatible with the Ore structure has $x \in J_c$ and $y \in J_\gamma$, and $\delta(x) = x^2$ forces $\gamma c = c^2$, that is, $\gamma = c$. Such a grading exists for every c of finite order, since $\delta(x^k) = kx^{k+1}$ has degree $c^{k+1} = \gamma c^k$. By Theorem B, a dual reflection group requires $\gamma = e$, hence $c = e$ and $G = \{e\}$. So the Jordan plane has no nontrivial compatible dual reflection group, although it has nontrivial compatible gradings. The hypothesis that G be a dual reflection group is therefore needed for the conclusion $\gamma = e$.

Once A is a graded subalgebra, a homogeneous Ore variable can be chosen after an affine change of variable.

Proposition 4.16. *Let $B = A[t;\sigma,\delta]$ be a graded Ore extension with $\deg t = 1$, and suppose that B has a G-grading for which A is a graded subalgebra. Then there is a homogeneous element $t' = \lambda t + a$ with $\lambda \in \Bbbk^\times$ and $a \in A_1$, and*

$$B = A[t';\sigma,\delta'], \quad \delta' = \lambda\delta + \delta_a. \tag{4.17}$$

In particular the grading is compatible with the Ore structure $A[t';\sigma,\delta']$, and δ' is outer if and only if δ is outer.

Proof. Since $B_1 = A_1 \oplus \Bbbk t$ and A_1 is a graded subspace of B_1, the quotient B_1/A_1 is a one-dimensional graded space, concentrated in some degree γ. An element of $B_1 \cap B_\gamma$ with nonzero image in B_1/A_1 has the form $t' = \lambda t + a$ with $\lambda \neq 0$ and $a \in A_1$. For $b \in A$,

$$t'b = \lambda\sigma(b)t + \lambda\delta(b) + ab = \sigma(b)t' + \lambda\delta(b) + \big(ab - \sigma(b)a\big) = \sigma(b)t' + (\lambda\delta + \delta_a)(b).$$

Expanding $(\lambda t + a)^n$ with (2.5) gives $t'^n \in \lambda^n t^n + \sum_{j<n} At^j$, so $B = \bigoplus_n At'^n$. This proves (4.17). Since $\delta' - \lambda\delta = \delta_a$ is inner, δ' is outer if and only if δ is. □

Example 4.18. The Ore variable need not be homogeneous even when A is graded. Let $B = \Bbbk[x][t] = \Bbbk[x,t]$, and grade it by $\mathbb{Z}_2 = \{e,\tau\}$ with $x \in B_\tau$ and $u = t + x \in B_e$. Since $B = \Bbbk[x,u]$, this is a grading of B, and $A = \Bbbk[x]$ is a graded subalgebra, but $t = u - x$ is not homogeneous. Proposition 4.16 replaces t by $t' = u$, with $\delta' = \delta_x = 0$, and the grading is compatible with $B = \Bbbk[x][u]$. Here $\gamma = e$, $B_e = \Bbbk[x^2,u]$, and $\mathbb{Z}_2$ is a dual reflection group for B.

The choice of t' in Proposition 4.16 is unique up to adding an element of $A_1 \cap A_\gamma$ and rescaling, which changes δ' by an inner σ-derivation. So the hypothesis $\delta \neq 0$ in Theorem B refers to a homogeneous choice of Ore variable, and it holds for every such choice when δ is outer. The remaining part of the compatibility assumption is that A be a graded subalgebra.

Remark 4.19. For the three-dimensional AS regular algebras of Ore type in [11, Table 1], graded by nonabelian groups, the grading is compatible with an Ore structure after a graded change of variables. By [11, Theorem 3.4] the grading corresponds to a quotient of the maximal grading group of the corresponding row of [11, Table 1], for which the basis u, v, z is homogeneous, and for the Ore-type rows this basis gives an Ore decomposition with A graded and z homogeneous. In dimension four, the algebra with a dual reflection group of order sixteen constructed in [19, §3.1] is a double Ore extension [48] of an AS regular algebra of dimension two [19, Proposition 3.1.1], but this presentation is not homogeneous for the grading, as noted after [19, Proposition 3.1.1]. So this example lies outside the scope of the present methods.

5. Iterated Ore extensions

In this section we prove Theorem C and describe the identity component as an iterated Ore extension. Let B be the iterated Ore extension (1.5), and fix a G-grading of B in which every x_i is homogeneous. As explained in Section 2, this grading is compatible with each Ore extension $B^{(i)} = B^{(i-1)}[x_i; \sigma_i, \delta_i]$. We write $c_i \in G$ for the G-degree of x_i, we put $r_i = |c_i|$ and $K_i = \mathrm{Supp}(B^{(i)})$, and we grade $B^{(i)}$ by K_i. Its identity component is $B_e^{(i)} = B^{(i)} \cap B_e$. Since B is a domain and the grading is faithful, $K_n = G$.

Lemma 5.1. *The following hold.*

(1) $K_1 = \langle c_1 \rangle$, *and for* $2 \leq i \leq n$ *the subgroup* K_{i-1} *is normal in* $K_i = K_{i-1}\langle c_i \rangle$.
(2) $G = \langle c_1 \rangle \langle c_2 \rangle \cdots \langle c_n \rangle$ *as a product of subsets, and*

$$|G| = \frac{r_1 r_2 \cdots r_n}{\prod_{i=2}^{n} |K_{i-1} \cap \langle c_i \rangle|} \leq r_1 r_2 \cdots r_n.$$

(3) *If* $\delta_i \neq 0$, *then* $c_i \in K_{i-1}$.
(4) *The equality* $|G| = r_1 \cdots r_n$ *holds if and only if* $K_{i-1} \cap \langle c_i \rangle = \{e\}$ *for* $2 \leq i \leq n$, *if and only if every* $g \in G$ *has a unique expression* $g = c_1^{a_1} \cdots c_n^{a_n}$ *with* $0 \leq a_i < r_i$.

Proof. The component $\Bbbk[x_1]_\beta$ is spanned by the powers x_1^k with $c_1^k = \beta$, so $K_1 = \langle c_1 \rangle$. For $i \geq 2$, Lemma 4.2(1),(2) applied to $B^{(i)} = B^{(i-1)}[x_i; \sigma_i, \delta_i]$ with its K_i-grading give (1), and Lemma 4.2(3) gives (3). By induction on i, (1) gives $K_i = \langle c_1 \rangle \cdots \langle c_i \rangle$ as a product of subsets, and $|K_i| = |K_{i-1}|\, r_i / |K_{i-1} \cap \langle c_i \rangle|$. Since $K_n = G$, this proves (2). The map $\prod_i \{0, \ldots, r_i - 1\} \to G$, $(a_i) \mapsto c_1^{a_1} \cdots c_n^{a_n}$, is surjective by (2), so it is bijective if and only if $|G| = r_1 \cdots r_n$, which by (2) happens if and only if every intersection $K_{i-1} \cap \langle c_i \rangle$ is trivial. □

Proof of Theorem C. By Lemma 5.1(2), the group G is the product of the cyclic subgroups $\langle c_i \rangle$, and the equivalence of (2) and (3) is Lemma 5.1(4).

Assume (1). We prove by descending induction on i that K_i is a dual reflection group for $B^{(i)}$ and that $K_{i-1} \cap \langle c_i \rangle = \{e\}$. For $i = n$ the first statement is the hypothesis. If K_i is a dual reflection group for $B^{(i)} = B^{(i-1)}[x_i; \sigma_i, \delta_i]$, then Theorem B shows that K_{i-1} is a dual reflection group for $B^{(i-1)}$ and that $K_{i-1} \cap \langle c_i \rangle = \{e\}$. Hence (2) holds by Lemma 5.1(4).

Assume (2). By Lemma 5.1(4), every intersection $K_{i-1} \cap \langle c_i \rangle$ is trivial. If $\delta_i \neq 0$, then $c_i \in K_{i-1}$ by Lemma 5.1(3), so $\langle c_i \rangle \subseteq K_{i-1} \cap \langle c_i \rangle$ and $c_i = e$. We prove by induction on i that K_i is a dual reflection group for $B^{(i)}$. For $i = 1$, the identity component of $\Bbbk[x_1]$ for its $\langle c_1 \rangle$-grading is $\Bbbk[x_1^{r_1}]$, which is AS regular. For $i \geq 2$, Theorem B applies to $B^{(i)} = B^{(i-1)}[x_i; \sigma_i, \delta_i]$, since K_{i-1} is a dual reflection group for $B^{(i-1)}$ and $K_{i-1} \cap \langle c_i \rangle = \{e\}$. For $i = n$ this is (1).

Assume now that (1), (2) and (3) hold. We saw that $c_i = e$ whenever $\delta_i \neq 0$. By Theorem B, $K_i = K_{i-1} \rtimes \langle c_i \rangle$, so the series

$$\{e\} \trianglelefteq K_1 \trianglelefteq K_2 \trianglelefteq \cdots \trianglelefteq K_n = G,$$

has cyclic factors $K_i / K_{i-1} \cong \langle c_i \rangle$, and G is solvable. Applying Proposition 4.6 to each step, with $f_{c_1^a} = x_1^a$ for $0 \leq a < r_1$ in $\Bbbk[x_1]$, we get by induction on i

$$f_{c_1^{a_1} \cdots c_i^{a_i}} = x_1^{a_1} \cdots x_i^{a_i} \quad (0 \leq a_j < r_j), \tag{5.2}$$

for the grading of $B^{(i)}$ by K_i. The step with $\delta_i \neq 0$ has $r_i = 1$ and does not change the elements f_g. By (4.8) and induction,

$$\begin{aligned} p_G(s) &= \prod_{i=1}^{n} [r_i]_s, \qquad m_G = c_1^{r_1-1} \cdots c_n^{r_n-1}, \\ N_G &= \sum_{i=1}^{n} (r_i - 1), \quad f_{m_G} = x_1^{r_1-1} \cdots x_n^{r_n-1}. \end{aligned} \tag{5.3}$$

Since $H_B(s) = (1-s)^{-n}$ by (2.9), the identity (2.19) gives

$$H_{B_e}(s) = \frac{1}{(1-s)^n \prod_i [r_i]_s} = \prod_{i=1}^{n} \frac{1}{1-s^{r_i}}.$$

Suppose finally that $(B_e)_1 = 0$. No x_i lies in B_e, so $c_i \neq e$ for all i, and therefore $\delta_i = 0$ for $i \geq 2$. By (2.20), each component $B_1 \cap B_g$ with $g \neq e$ is at most one-dimensional, and $B_1 \cap B_e = 0$. Since $x_1, \ldots, x_n$ are homogeneous and linearly independent, their G-degrees are pairwise distinct, $B_1 \cap B_{c_j} = \Bbbk x_j$, and the other components of B_1 vanish. Let $j < i$. Since $(\sigma_i, 0)$ is c_i-compatible, $\sigma_i(x_j)$ is a nonzero element of $(B^{(i-1)})_1 \cap B_{c_i c_j c_i^{-1}}$, so $c_i c_j c_i^{-1} = c_{\pi_i(j)}$ for some $\pi_i(j) < i$, and $\sigma_i(x_j) = \lambda_{ij} x_{\pi_i(j)}$ with $\lambda_{ij} \neq 0$. Since σ_i is injective on $(B^{(i-1)})_1$, the map π_i is a permutation of $\{1, \ldots, i-1\}$. The binomial presentation follows from (2.11). □

Theorem C turns the search for compatible dual reflection groups into a problem about finite quotients of one finitely presented group. A grading of B with $x_i \in B_{c_i}$ exists if and only if the relations (2.11) are homogeneous. Since the monomials $x_1^{a_1} \cdots x_n^{a_n}$ are linearly independent in B, this means that $\sigma_i(x_j)$ is a combination of the x_k with $c_k = c_i c_j c_i^{-1}$, and that $\delta_i(x_j)$, written in the monomial basis, is a combination of monomials $x_k x_l$ with $k \leq l$ and $c_k c_l = c_i c_j$. Let Γ_B be the group generated by symbols $\hat{c}_1, \ldots, \hat{c}_n$ subject to the relations

$$\hat{c}_k = \hat{c}_i \hat{c}_j \hat{c}_i^{-1}, \quad \hat{c}_k \hat{c}_l = \hat{c}_i \hat{c}_j,$$

for all $1 \leq j < i \leq n$, all variables x_k occurring in $\sigma_i(x_j)$ and all monomials $x_k x_l$ occurring in $\delta_i(x_j)$ with nonzero coefficient. Then the gradings of B in which every x_i is homogeneous are exactly the gradings induced by the surjections $\Gamma_B \to G$ onto finite groups. By Theorem C, the dual reflection groups among them are exactly the quotients in which the order of G is the product of the orders of the images of the $\hat{c}_i$.

Example 5.4. Let $n \geq 2$, let $A = \Bbbk[x_1, \ldots, x_n]$, and let $\delta = x_1 E$, where $E = \sum_i x_i \frac{\partial}{\partial x_i}$ is the Euler derivation. So $\delta(x_1) = x_1^2$ and $\delta(x_i) = x_1 x_i$, and δ raises degrees by one. The Ore extension

$$B = A[t; \delta] = \Bbbk\langle x_1, \ldots, x_n, t \rangle / \big(x_i x_j - x_j x_i,\ tx_1 - x_1 t - x_1^2,\ tx_i - x_i t - x_1 x_i \ (i \geq 2) \big)$$

is AS regular of dimension $n+1$, and t is not normal, since $Bt = \bigoplus_{k \geq 1} At^k$ does not contain $tx_1 = x_1 t + x_1^2$. The Jordan plane is the analogous algebra with $n = 1$. Let $r_2, \ldots, r_n \geq 1$ and grade B by $G = \prod_{i \geq 2} \mathbb{Z}_{r_i}$, with x_1 and t of degree e and x_i of degree a generator c_i of the factor $\mathbb{Z}_{r_i}$. The relations are homogeneous since $\delta(x_i) = x_1 x_i$ has degree c_i. Since $|G| = \prod_{i \geq 2} r_i$, Theorem C shows that G is a dual reflection group for B, with $x_1, t \in B_e$, and by Theorem B

$$B_e = \Bbbk[x_1, y_2, \ldots, y_n][t; \delta_e], \quad y_i = x_i^{r_i}, \quad \delta_e(x_1) = x_1^2, \quad \delta_e(y_i) = r_i x_1 y_i.$$

Here $x_1 \in (A_e)_1$, so $(A_e)_1 \neq 0$, as predicted by Theorem B, and δ induces the zero map on $C(A) = \bigotimes_{i \geq 2} \Bbbk[\bar{x}_i]/(\bar{x}_i^{r_i})$, as in Proposition 4.9, since $\delta(x_i) = x_1 x_i \in A(A_e)_{\geq 1}$. By Theorem D, $C(B) \cong C(A)$ and $\operatorname{gci}(C(B)) = \#\{i \geq 2 \mid r_i \geq 2\}$. Moreover $f_{m_G} = x_2^{r_2-1} \cdots x_n^{r_n-1}$ and $\ell_{B_e} = 2 + \sum_{i \geq 2} r_i$.

Remark 5.5. By [27, Theorem 0.3(3)], $\deg f_g$ is the length of g with respect to the set of degrees of the homogeneous elements of $B_1 \setminus B_e$. Here this set is $\{c_i \mid c_i \neq e\}$, and (5.2) shows that $\deg f_g = a_1 + \cdots + a_n$. So the normal form of Theorem C(3) is a word of minimal length representing g.

The proof also describes the identity component as an iterated Ore extension.

Proposition 5.6. *In the setting of Theorem C, suppose that G is a dual reflection group for B, and put $y_i = x_i^{r_i}$. Then $B_e^{(i)}$ is generated by $y_1, \ldots, y_i$, and*

$$B_e = \Bbbk[y_1][y_2; \sigma_2', \delta_2'] \cdots [y_n; \sigma_n', \delta_n'], \quad \deg y_i = r_i, \tag{5.7}$$

where $\sigma_i' = \sigma_i^{r_i}|_{B_e^{(i-1)}}$ *and* $\delta_i' = \delta_i|_{B_e^{(i-1)}}$*, with* $\delta_i = 0$ *whenever* $r_i > 1$*. Moreover*

$$\ell_{B_e} = r_1 + r_2 + \cdots + r_n = \ell_B + N_G. \tag{5.8}$$

Proof. By the proof of Theorem C, K_i is a dual reflection group for $B^{(i)}$ and $K_{i-1} \cap \langle c_i \rangle = \{e\}$. So (4.5) gives $B_e^{(i)} = B_e^{(i-1)}[y_i; \sigma_i', \delta_i']$. Since $B_e^{(1)} = \Bbbk[y_1]$, induction on i gives (5.7). Since $\Bbbk$ has Gorenstein parameter 0, Corollary 3.11 gives $\ell_{B_e^{(i)}} = \ell_{B_e^{(i-1)}} + r_i$, hence $\ell_{B_e} = \sum_i r_i$, and $\ell_B = n$ and $N_G = \sum_i (r_i - 1)$ by (5.3). □

The identity (5.8) is an instance of [27, Lemma 3.4(3)], since the Frobenius algebra $C(B)$ has Gorenstein parameter $-N_G$.

Corollary 5.9. *Let* (q_{ij}) *be a multiplicatively antisymmetric matrix in* $\Bbbk^\times$*, that is,* $q_{ii} = 1$ *and* $q_{ji} = q_{ij}^{-1}$*, and let*

$$\mathcal{O}_q(\Bbbk^n) = \Bbbk\langle x_1, \ldots, x_n \rangle / \big(x_i x_j - q_{ij} x_j x_i\big) \tag{5.10}$$

be the quantum affine space, viewed as the iterated Ore extension with $\sigma_i(x_j) = q_{ij} x_j$ *for* $j < i$ *and* $\delta_i = 0$*. Let* G *be a finite group with a grading of* $\mathcal{O}_q(\Bbbk^n)$ *in which every* x_i *is homogeneous, of degree* c_i*. Then the* c_i *commute, and* G *is a dual reflection group if and only if the multiplication map* $\prod_i \langle c_i \rangle \to G$ *is an isomorphism. In this case*

$$B_e \cong \mathcal{O}_{q'}(\Bbbk^n), \quad q_{ij}' = q_{ij}^{r_i r_j},$$

on the generators $x_1^{r_1}, \ldots, x_n^{r_n}$*.*

Proof. Since $\sigma_i(x_j) = q_{ij} x_j$ is a nonzero element of $B_{c_i c_j c_i^{-1}} \cap B_{c_j}$, the elements c_i and c_j commute. So $G = \langle c_1 \rangle \cdots \langle c_n \rangle$ is abelian, the multiplication map $\prod_i \langle c_i \rangle \to G$ is a surjective homomorphism, and it is an isomorphism if and only if $|G| = \prod_i r_i$. The first statement is now Theorem C. By Proposition 5.6, B_e is generated by the elements $x_i^{r_i}$, and the relations (5.10) give, by induction on the exponents,

$$x_i^a x_j^b = q_{ij}^{ab}\, x_j^b x_i^a, \quad \text{so} \quad x_i^{r_i} x_j^{r_j} = q_{ij}^{r_i r_j} x_j^{r_j} x_i^{r_i}.$$

The monomials in the $x_i^{r_i}$ are linearly independent, so these are the defining relations of B_e. □

In the dual language, the G-grading of $\mathcal{O}_q(\Bbbk^n)$ is a diagonal action of the character group $G^\vee$, and Corollary 5.9 is the diagonal case of the theorem of Kirkman, Kuzmanovich and Zhang for skew polynomial rings [24, Theorem 4.5].

In dimension two, Theorem C applies to the quantum plane and the Jordan plane. For $\Bbbk_q[x, y] = \Bbbk[x][y; \sigma]$, the G-degrees c_1 of x and c_2 of y commute for every compatible grading, and G is a dual reflection group if and only if $\langle c_1 \rangle \cap \langle c_2 \rangle = \{e\}$. Then $G = \mathbb{Z}_{r_1} \times \mathbb{Z}_{r_2}$, $B_e = \Bbbk_{q^{r_1 r_2}}[x^{r_1}, y^{r_2}]$ and $f_{m_G} = x^{r_1 - 1} y^{r_2 - 1}$. For the Jordan plane, the only compatible dual reflection group is trivial (Example 4.15). These gradings are abelian. For the quantum plane they are among the coactions classified by Buzaglo and Rogalski [10, Theorem 1.2].

Example 5.11. We describe the three-dimensional case with $(B_e)_1 = 0$. Let B be an iterated Ore extension (1.5) with $n = 3$, and let G be a dual reflection group for B with every x_i homogeneous and $(B_e)_1 = 0$. By Theorem C, $\delta_2 = \delta_3 = 0$, the degrees c_1, c_2, c_3 are distinct and different from e, and the relations are binomial. The permutation π_2 of $\{1\}$ is trivial, so $x_2 x_1 = q x_1 x_2$ for some $q \in \Bbbk^\times$, and c_1, c_2 commute. If π_3 is the identity, then G is abelian and B is a quantum affine space as in Corollary 5.9. Otherwise π_3 is the transposition, and

$$B = \Bbbk\langle x_1, x_2, x_3 \rangle / \big(x_2 x_1 - q x_1 x_2,\ x_3 x_1 - \lambda x_2 x_3,\ x_3 x_2 - \mu x_1 x_3\big) \quad (\lambda, \mu \in \Bbbk^\times). \tag{5.12}$$

The assignment $\sigma_3(x_1) = \lambda x_2$, $\sigma_3(x_2) = \mu x_1$ defines an endomorphism of $\Bbbk_q[x_1, x_2]$ if and only if

$$\sigma_3(x_2)\sigma_3(x_1) - q\sigma_3(x_1)\sigma_3(x_2) = \lambda\mu\,(x_1 x_2 - q x_2 x_1) = \lambda\mu(1 - q^2)\, x_1 x_2$$

vanishes, that is, if and only if $q = \pm 1$. Conjugation by c_3 exchanges c_1 and c_2, so $r_1 = r_2 =: r \geq 2$, conjugation by c_3 is an automorphism of order two of $\langle c_1 \rangle \times \langle c_2 \rangle$, and $r_3 = 2m$ is even. By Theorem C,

$$G = \big(\langle c_1 \rangle \times \langle c_2 \rangle\big) \rtimes \langle c_3 \rangle \cong (\mathbb{Z}_r \times \mathbb{Z}_r) \rtimes \mathbb{Z}_{2m},$$

where the generator of $\mathbb{Z}_{2m}$ exchanges the two factors, and c_3^2 is central. Conversely, for $q = \pm 1$, $\lambda, \mu \in \Bbbk^\times$, $r \geq 2$ and $m \geq 1$, the relations (5.12) are homogeneous for the grading of B by this group with $x_i \in B_{c_i}$ for

$i = 1, 2, 3$, because $c_3c_1 = c_2c_3$ and $c_3c_2 = c_1c_3$. Since $|G| = r \cdot r \cdot 2m$, Theorem C shows that G is a dual reflection group, and Proposition 5.6 gives

$$B_e = \Bbbk\langle y_1, y_2, y_3\rangle/\big(y_2y_1 - q^{r^2}y_1y_2,\ y_3y_1 - (\lambda\mu)^{mr}y_1y_3,\ y_3y_2 - (\lambda\mu)^{mr}y_2y_3\big),$$

with $y_1 = x_1^r$, $y_2 = x_2^r$ and $y_3 = x_3^{2m}$, since $\sigma_3^{2m}(x_j) = (\lambda\mu)^m x_j$ for $j = 1, 2$. This recovers, for the algebras of Ore type, the family of groups $(\mathbb{Z}_r \times \mathbb{Z}_r) \rtimes \mathbb{Z}_{2m}$ in the classification of Buzaglo and Rogalski [11, Theorem 1.2], together with an explicit identity component.

The following family was constructed by Goetz, Kirkman, Moore and Vashaw [19, Example 2.1.7], who also allow $\Bbbk_{-1}[x_1, \dots, x_n]$ in place of $\Bbbk[x_1, \dots, x_n]$. We treat the commutative case, and the same argument applies to the skew case. Its dual reflection groups follow from Theorems B and C.

Example 5.13. Let $n \geq 2$, $r \geq 1$ and $p \geq 1$. Let $A = \Bbbk[x_1, \dots, x_n]$, let σ be the automorphism with $\sigma(x_i) = x_{i+1}$, indices modulo n, and put

$$B = A[t; \sigma] = \Bbbk\langle x_1, \dots, x_n, t\rangle/\big(x_ix_j - x_jx_i,\ tx_i - x_{i+1}t\big). \tag{5.14}$$

It is the iterated Ore extension $\Bbbk[x_1][x_2]\cdots[x_n][t; \sigma]$, AS regular of dimension $n + 1$. Let $\varepsilon_1, \dots, \varepsilon_n$ be generators of $\mathbb{Z}_r^n = \langle\varepsilon_1\rangle \times \cdots \times \langle\varepsilon_n\rangle$, written multiplicatively, and let

$$G = \mathbb{Z}_r^n \rtimes \langle\gamma\rangle, \quad |\gamma| = np, \quad \gamma\varepsilon_i\gamma^{-1} = \varepsilon_{i+1}. \tag{5.15}$$

For $p = 1$ this is the wreath product $\mathbb{Z}_r \wr \mathbb{Z}_n$. Grade B by $x_i \in B_{\varepsilon_i}$ and $t \in B_\gamma$. The automorphism σ maps a monomial of G-degree $g = \varepsilon_1^{a_1}\cdots\varepsilon_n^{a_n}$ to one of G-degree $\varepsilon_2^{a_1}\cdots\varepsilon_1^{a_n} = \gamma g\gamma^{-1}$, so the relations are homogeneous. Here $K = \mathbb{Z}_r^n$ is a dual reflection group for A with $A_e = \Bbbk[x_1^r, \dots, x_n^r]$, and $K \cap \langle\gamma\rangle = \{e\}$. Equivalently, $|G| = r^n \cdot np = |\varepsilon_1|\cdots|\varepsilon_n|\,|\gamma|$. By Theorem B, G is a dual reflection group for B, with

$$B_e = \Bbbk[x_1^r, \dots, x_n^r][t^{np}; \sigma^{np}] = \Bbbk[x_1^r, \dots, x_n^r, t^{np}],$$

since $\sigma^n = \mathrm{id}$. By Theorem C and Proposition 5.6,

$$p_G(s) = [r]_s^n\,[np]_s, \quad |G| = npr^n, \quad f_{m_G} = (x_1\cdots x_n)^{r-1}t^{np-1}, \quad \ell_{B_e} = nr + np.$$

The group G is nonabelian if and only if $r \geq 2$. When $r \geq 2$ we have $(B_e)_1 = 0$, and the defining relations (5.14) are binomial, as predicted by Theorem C.

Corollary 5.16. *Let $B = \Bbbk[x_1, \dots, x_n][t; \sigma]$ be as in Example 5.13, and let G be a finite group with a grading of B in which $x_1, \dots, x_n$ and t are homogeneous. Then G is a dual reflection group for B if and only if either $x_1, \dots, x_n \in B_e$ and G is cyclic of arbitrary order, generated by the G-degree of t, or $G \cong \mathbb{Z}_r^n \rtimes \mathbb{Z}_{np}$ as in* (5.15) *for some $r \geq 2$ and $p \geq 1$, with the grading of Example 5.13.*

Proof. Let ε_i and γ be the G-degrees of x_i and t. The relations $x_ix_j = x_jx_i$ and $tx_i = x_{i+1}t$ are homogeneous exactly when the ε_i commute and $\gamma\varepsilon_i\gamma^{-1} = \varepsilon_{i+1}$. So the ε_i are conjugate and have a common order r, and $E = \langle\varepsilon_1, \dots, \varepsilon_n\rangle$ is abelian of exponent dividing r, with $|E| \leq r^n$. By Lemma 4.2(2), $G = E\langle\gamma\rangle$ with E normal, so

$$|G| \leq |E|\,|\gamma| \leq r^n|\gamma|,$$

with equality if and only if $E \cong \mathbb{Z}_r^n$ with basis $\varepsilon_1, \dots, \varepsilon_n$ and $E \cap \langle\gamma\rangle = \{e\}$. By Theorem C, G is a dual reflection group exactly in this case. If $r = 1$, then $G = \langle\gamma\rangle$. If $r \geq 2$, the elements $\varepsilon_1, \dots, \varepsilon_n$ are pairwise distinct, and $\gamma^{|\gamma|} = e$ gives $\varepsilon_{i+|\gamma|} = \varepsilon_i$, so n divides $|\gamma|$ and G is the group (5.15) with $np = |\gamma|$. Conversely, the groups with $r \geq 2$ are dual reflection groups by Example 5.13. If $r = 1$ and $|\gamma| = N$, then $K = \{e\}$, and Theorem B shows that $\langle\gamma\rangle$ is a dual reflection group, with $B_e = A[t^N; \sigma^N]$. □

Example 5.17. We make Example 5.13 explicit for $n = r = 2$ and $p = 1$. Here $B = \Bbbk[x_1, x_2][t; \sigma]$ with $\sigma(x_1) = x_2$ and $\sigma(x_2) = x_1$, that is,

$$x_2x_1 = x_1x_2, \quad tx_1 = x_2t, \quad tx_2 = x_1t. \tag{5.18}$$

These are the binomial relations of Theorem C, with π_3 the transposition of $\{1, 2\}$. The group is $G = (\langle\varepsilon_1\rangle \times \langle\varepsilon_2\rangle) \rtimes \langle\gamma\rangle$ with $\varepsilon_1^2 = \varepsilon_2^2 = \gamma^2 = e$ and $\gamma\varepsilon_1\gamma^{-1} = \varepsilon_2$. The element $\rho = \varepsilon_1\gamma$ satisfies $\rho^2 = \varepsilon_1\gamma\varepsilon_1\gamma^{-1} = \varepsilon_1\varepsilon_2$,

so it has order four, and $\gamma\rho\gamma^{-1} = \varepsilon_2\gamma = \rho^{-1}$. Since $\gamma \notin \langle\rho\rangle$, the group $G = \langle\rho\rangle \rtimes \langle\gamma\rangle$ is the dihedral group of order eight. By (5.2) the elements f_g are

g	e	ε_1	ε_2	$\varepsilon_1\varepsilon_2$	γ	$\varepsilon_1\gamma$	$\varepsilon_2\gamma$	$\varepsilon_1\varepsilon_2\gamma$
f_g	1	x_1	x_2	x_1x_2	t	x_1t	x_2t	x_1x_2t
$\deg f_g$	0	1	1	2	1	2	2	3

So $p_G(s) = (1+s)^3$, $m_G = \varepsilon_1\varepsilon_2\gamma$ and $f_{m_G} = x_1x_2t$. The identity component $B_e = \Bbbk[x_1^2, x_2^2, t^2]$ is a polynomial ring with Hilbert series $(1-s^2)^{-3}$ and Gorenstein parameter $6 = \ell_B + \deg f_{m_G}$. The identity (2.17) can be checked directly. For instance $f_{\varepsilon_1} f_{\varepsilon_1^{-1}m_G} = x_1 \cdot x_2t = f_{m_G}$ and $f_\gamma f_{\gamma^{-1}m_G} = t \cdot x_1x_2 = x_2x_1t = f_{m_G}$, since $\gamma^{-1}\varepsilon_1\varepsilon_2\gamma = \varepsilon_1\varepsilon_2$.

6. Covariant rings and complete intersections

In this section we determine the structure of the covariant rings of iterated Ore extensions and the growth of their Ext-algebras, and we prove Theorem D.

Proposition 6.1. *In the setting of Theorem B, suppose that G is a dual reflection group for B. Then σ induces a graded automorphism $\bar\sigma$ of $C(A)$, and the inclusion $A \subseteq B$ induces an isomorphism of graded algebras*

$$C(B) \cong C(A)[\bar t; \bar\sigma]/(\bar t^{\,r}).$$

In particular $C(B) \cong C(A)$ if $\delta \neq 0$.

Proof. Put $J = A(A_e)_{\geq 1}$, a two-sided ideal of A by (2.16). Since $\sigma(A_e) = A_e$, we have $\sigma(J) = J$, so σ induces $\bar\sigma$ on $C(A) = A/J$. By (4.5), $(B_e)_{\geq 1} = (A_e)_{\geq 1} \oplus \bigoplus_{l\geq 1} A_e t^{rl}$, so

$$B(B_e)_{\geq 1} = B(A_e)_{\geq 1} + Bt^r, \quad Bt^r = \bigoplus_{n\geq r} At^n.$$

We claim that

$$B(B_e)_{\geq 1} = \bigoplus_{n<r} Jt^n \oplus \bigoplus_{n\geq r} At^n. \tag{6.2}$$

The right side is contained in the left side, since $Jt^n = A(A_e)_{\geq 1}t^n$ and $(A_e)_{\geq 1}t^n = t^n\sigma^{-n}\big((A_e)_{\geq 1}\big)$ when $\delta = 0$, while $r = 1$ when $\delta \neq 0$. For the reverse inclusion it suffices to show that At^nb lies in the right side for $b \in (A_e)_{\geq 1}$ and $n \geq 0$. If $\delta = 0$, then $At^nb = A\sigma^n(b)t^n \subseteq Jt^n$. If $\delta \neq 0$, then $r = 1$, and (2.6) gives $t^nb \in \delta^n(b) + Bt$ for $n \geq 1$, with $\delta^n(b) \in (A_e)_{\geq 1}$ since δ preserves A_e and raises degrees. So $At^nb \subseteq Jt^n$ if $\delta = 0$, and $At^nb \subseteq J + Bt$ if $\delta \neq 0$, where $r = 1$. In both cases At^nb lies in the right side of (6.2), which proves it. The map $B = \bigoplus_n At^n \to C(A)[\bar t; \bar\sigma]/(\bar t^{\,r})$, $at^n \mapsto \bar a\bar t^n$, is a surjective homomorphism of graded algebras by the universal property of Ore extensions, because $\bar\delta = 0$ on $C(A)$ when $\delta \neq 0$ by Proposition 4.9. Its kernel is the right side of (6.2). □

For example, in Example 4.14 we have $\delta \neq 0$, and

$$B(B_e)_{\geq 1} = A(A_e)_{\geq 1} \oplus Bz, \quad A(A_e)_{\geq 1} = Ax + Ay^2, \quad C(B) \cong C(A) = \Bbbk[\bar y]/(\bar y^{\,2}),$$

with Hilbert series $1 + s = p_G(s)$. The Ore variable z and the generator x of A_e both disappear in the covariant ring.

Let C' be a finite-dimensional connected graded algebra, let $\bar\sigma$ be a graded automorphism of C', let $r \geq 2$, and put

$$C = C'[x; \bar\sigma]/(x^r), \quad \deg x = 1.$$

The element x is normal in $C'[x; \bar\sigma]$, so $(x^r) = x^rC'[x; \bar\sigma]$, and $C = \bigoplus_{j<r} C'x^j = \bigoplus_{j<r} x^jC'$ is free as a left and as a right C'-module. The ideal $xC = Cx$ is two-sided with $C/xC \cong C'$, and we write $\pi\colon C \to C'$ for the quotient map. Let $F_\bullet \to \Bbbk$ be a minimal graded free resolution of $\Bbbk_{C'}$. Since C' is finite-dimensional, each F_p has finite rank, and $\operatorname{Ext}^p_{C'}(\Bbbk, \Bbbk) \cong \operatorname{Hom}_{C'}(F_p, \Bbbk)$.

Let $F^{\bar\sigma}$ be the complex $F_\bullet$ with the right action $f * c = f\bar\sigma(c)$. It is a free resolution of $\Bbbk^{\bar\sigma} = \Bbbk$, so the identity of $\Bbbk$ lifts to a C'-linear chain map $F \to F^{\bar\sigma}$, that is, to a degree-preserving chain map $\theta\colon F_\bullet \to F_\bullet$ with

$$\theta(fc) = \theta(f)\,\bar\sigma(c) \quad (f \in F_p,\ c \in C'), \tag{6.3}$$

and $\theta_0(1) = 1$ on $F_0 = C'$. For $k \geq 1$, the map

$$\varphi_k \colon F_p \otimes_{C'} C \to F_p \otimes_{C'} C, \quad \varphi_k(f \otimes c) = \theta^k(f) \otimes x^k c,$$

is well defined, since $\theta^k(fc') \otimes x^k c = \theta^k(f) \otimes \bar{\sigma}^k(c') x^k c = \theta^k(f) \otimes x^k c' c$ for $c' \in C'$. It is right C-linear, it raises the internal degree by k, and it commutes with $d_F \otimes 1$. Moreover $\varphi_1 \varphi_{r-1} = \varphi_{r-1} \varphi_1 = \varphi_r = 0$, since $x^r = 0$ in C.

Proposition 6.4. *For $p, q \geq 0$, put $D_{p,q} = F_p \otimes_{C'} C(-s_q)$, where $s_{2j} = jr$ and $s_{2j+1} = jr + 1$. Let the horizontal differential $D_{p,q} \to D_{p-1,q}$ be $d_F \otimes 1$, and let the vertical differential $D_{p,q} \to D_{p,q-1}$ be φ_1 for q odd and φ_{r-1} for $q \geq 2$ even.*

(1) *The total complex $\operatorname{Tot} D$, with differential $d_F \otimes 1 + (-1)^p \varphi$ on $D_{p,q}$, is a minimal graded free resolution of $\Bbbk_C$.*

(2) *The Poincaré series of C and C' are related by*

$$\begin{aligned} P_C(y,s) &= P_{C'}(y,s)\, \frac{1+ys}{1-y^2 s^r}, \\ \dim \operatorname{Ext}^n_C(\Bbbk,\Bbbk) &= \sum_{p=0}^{n} \dim \operatorname{Ext}^p_{C'}(\Bbbk,\Bbbk). \end{aligned} \tag{6.5}$$

(3) *The algebra map $\pi^* \colon E(C') \to E(C)$ induced by π is injective. There is $\eta \in \operatorname{Ext}^2_C(\Bbbk,\Bbbk)_{-r}$ such that the linear map*

$$E(C') \otimes \Bbbk[\eta] \to E(C), \quad a \otimes \eta^j \mapsto \pi^*(a)\, \eta^j,$$

is injective, with image $\bigoplus_{j \geq 0} \operatorname{Hom}_C(\bigoplus_p D_{p,2j}, \Bbbk)$.

Proof. (1) The vertical maps commute with the horizontal ones and compose to zero, so D is a double complex concentrated in the first quadrant, and the sign makes $\operatorname{Tot} D$ a complex. Since C is free as a left C'-module, each row $D_{\bullet,q}$ is a free resolution of $N(-s_q)$, where

$$N = \Bbbk \otimes_{C'} C = C/C'_+ C \cong \Bbbk[\bar{x}]/(\bar{x}^r).$$

Here $C'_+ C = C C'_+$ is a two-sided ideal, since $x C'_+ = \bar{\sigma}(C'_+) x = C'_+ x$. On $F_0 \otimes_{C'} C = C$ we have $\varphi_k(c) = x^k c$, since $\theta_0 = \bar{\sigma}$. So the induced maps on H_0 of the rows are left multiplication by $\bar{x}$ and by $\bar{x}^{r-1}$, and they form the complex

$$\cdots \to N(-r-1) \xrightarrow{\bar{x}} N(-r) \xrightarrow{\bar{x}^{r-1}} N(-1) \xrightarrow{\bar{x}} N \to \Bbbk \to 0.$$

It is exact, since in $\Bbbk[\bar{x}]/(\bar{x}^r)$ the kernel of multiplication by $\bar{x}$ is $\bar{x}^{r-1} N$ and the kernel of multiplication by $\bar{x}^{r-1}$ is $\bar{x} N$. A first-quadrant double complex whose rows are resolutions has total complex quasi-isomorphic to the complex of the H_0 of the rows, so $\operatorname{Tot} D$ is a free resolution of $\Bbbk_C$. It is minimal, because $d_F \otimes 1$ has entries in C'_+ by minimality of F, while φ_1 and φ_{r-1} have image in $(\operatorname{Tot} D)\, xC$ and $(\operatorname{Tot} D)\, x^{r-1} C$, which lie in $(\operatorname{Tot} D)\, C_+$ since $r \geq 2$.

(2) By (1), $\operatorname{Ext}^n_C(\Bbbk,\Bbbk) = \operatorname{Hom}_C(\operatorname{Tot}_n D, \Bbbk) = \bigoplus_{p+q=n} \operatorname{Hom}_C(D_{p,q}, \Bbbk)$, and

$$\operatorname{Hom}_C(D_{p,q}, \Bbbk) \cong \operatorname{Hom}_{C'}(F_p, \Bbbk)(s_q) \cong \operatorname{Ext}^p_{C'}(\Bbbk,\Bbbk)(s_q).$$

Hence $P_C(y,s) = P_{C'}(y,s) \sum_{q \geq 0} y^q s^{s_q}$, and $\sum_q y^q s^{s_q} = (1+ys) \sum_j y^{2j} s^{jr} = (1+ys)/(1-y^2 s^r)$. Setting $s = 1$ gives the second formula of (6.5), since $(1+y)/(1-y^2) = (1-y)^{-1}$.

(3) The maps $f \otimes c \mapsto f\pi(c)$ on $D_{p,0}$, and zero on $D_{p,q}$ for $q > 0$, form a chain map $\Phi \colon \operatorname{Tot} D \to F$ over π lifting the identity of $\Bbbk$. Indeed, $\Phi\varphi_1(f \otimes c) = \theta(f)\pi(xc) = 0$. So $\pi^*(a)$ is represented by the cocycle $a \circ \Phi$, which is a on $D_{p,0}$ and zero elsewhere, and π^* is injective. The identity maps $D_{p,q+2} \to D_{p,q}(-r)$ form a chain map $\Theta \colon \operatorname{Tot} D \to \operatorname{Tot} D$ of homological degree -2, because the vertical differentials are 2-periodic. Its component $D_{0,2} = C(-r) \to D_{0,0}(-r) = C(-r)$ is the identity, so Θ lifts the cocycle dual to the generator of $D_{0,2}$, that is, the class $\eta \in \operatorname{Ext}^2_C(\Bbbk,\Bbbk)$. By the chain-map description of the Yoneda product, Θ^j lifts η^j, and the Yoneda product $\pi^*(a)\eta^j$ is represented by $a \circ \Phi \circ \Theta^j$. This cocycle is a on $D_{p,2j}$ and zero elsewhere. As a runs through a basis of $\operatorname{Ext}^p_{C'}(\Bbbk,\Bbbk)$ and $j \geq 0$, these cocycles are linearly independent, which proves (3). □

Example 6.6. Let $C' = \Bbbk[a]/(a^2)$ with $\bar{\sigma}(a) = -a$, and $r = 3$, so that

$$C = \Bbbk\langle a, t\rangle/(a^2,\ ta + at,\ t^3), \quad H_C(s) = (1+s)(1+s+s^2).$$

The minimal resolution of $\Bbbk_{C'}$ has $F_p = e_p C'$ with e_p of degree p and $d(e_p) = e_{p-1}a$. The chain map θ of (6.3) is $\theta(e_p c) = (-1)^p e_p \bar{\sigma}(c)$, since $d\theta(e_p) = \theta(e_{p-1}a) = \theta(e_{p-1})\bar{\sigma}(a)$ forces the signs to alternate. So $\operatorname{Tot}_n D$ is free on generators $e_{p,q}$ with $p+q=n$, of internal degree $p+s_q$, and

$$d(e_{p,q}) = e_{p-1,q}\, a + (-1)^{p+pk_q}\, e_{p,q-1}\, t^{k_q}, \quad k_q = \begin{cases} 1 & q \text{ odd}, \\ 2 & q \geq 2 \text{ even}, \end{cases}$$

with the convention $e_{-1,q} = e_{p,-1} = 0$. For instance $d(e_{1,1}) = e_{0,1}a + e_{1,0}t$, and $d^2(e_{1,1}) = e_{0,0}(ta + at) = 0$. Here $P_{C'}(y,s) = (1-ys)^{-1}$, and (6.5) gives

$$P_C(y,s) = \frac{1+ys}{(1-ys)(1-y^2s^3)}, \quad \dim \operatorname{Ext}^n_C(\Bbbk,\Bbbk) = n+1,$$

so $\operatorname{gci}(C) = 2$. This is the covariant ring of the dual reflection group $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$ of the quantum plane $\Bbbk_{-1}[x_1,x_2] = \Bbbk[x_1][x_2;\sigma]$, $\sigma(x_1) = -x_1$, with G-degrees of x_1 and x_2 of orders two and three, where $a = \bar{x}_1$ and $t = \bar{x}_2$.

The resolution of Proposition 6.4(1) is the graded case $\delta = 0$ of the resolution of McPhate for truncated Ore extensions [34, Theorem 3.10], which is built from the twisted product resolutions of Shepler and Witherspoon [42, Theorem 5.12]. We need its minimality and the explicit form of Proposition 6.4(3) to control the growth of $E(C)$.

Proof of Theorem D. For $0 \leq i \leq n$, let $C^{(i)}$ be the covariant ring of $B^{(i)}$ for its K_i-grading, with $C^{(0)} = \Bbbk$. By the proof of Theorem C, K_i is a dual reflection group for $B^{(i)} = B^{(i-1)}[x_i;\sigma_i,\delta_i]$ and $K_{i-1} \cap \langle c_i \rangle = \{e\}$, and $c_i = e$ whenever $\delta_i \neq 0$. So Proposition 6.1 gives

$$C^{(i)} \cong C^{(i-1)}[\bar{x}_i;\bar{\sigma}_i]/(\bar{x}_i^{r_i}),$$

which is $C^{(i-1)}$ when $c_i = e$. Since $C^{(n)} = C(B)$, this proves the description of $C(B)$. Let $m = \#\{i \mid c_i \neq e\}$. Applying Proposition 6.4(2) at each step with $r_i \geq 2$, starting from $P_\Bbbk = 1$, gives

$$P_{C(B)}(y,s) = \prod_{c_i \neq e} \frac{1+ys}{1-y^2s^{r_i}}, \quad \dim \operatorname{Ext}^n_{C(B)}(\Bbbk,\Bbbk) = \binom{n+m-1}{m-1} \quad (m \geq 1).$$

If $m = 0$, then every c_i is trivial, so $G = \{e\}$, $C(B) = \Bbbk$ and $\operatorname{gci}(C(B)) = 0$. Assume now that $m \geq 1$. It remains to show that $\operatorname{GKdim} E(C(B)) = m$. Write $E = E(C(B))$. Let $W \subseteq E$ be a finite-dimensional subspace containing 1, contained in $\bigoplus_{n \leq d} E^n$. Then $W^N \subseteq \bigoplus_{n \leq dN} E^n$ has dimension at most $\sum_{n \leq dN} (n+1)^{m-1} \leq (dN+1)^m$, so $\operatorname{GKdim} E \leq m$. For the reverse inequality, let $i_1 < \cdots < i_m$ be the indices with $c_i \neq e$, and let $\pi_k^* \colon E(C^{(i_k-1)}) \to E(C^{(i_k)})$ and $\eta_k \in E^2(C^{(i_k)})$ be as in Proposition 6.4(3), where $C^{(i_k-1)} = C^{(i_{k-1})}$. Put $W_0 = \Bbbk$ and $W_k = \pi_k^*(W_{k-1}) + \Bbbk\eta_k$. By Proposition 6.4(3), W_k^{2N} contains the linearly independent families $\pi_k^*(W_{k-1}^N)\eta_k^j$ for $0 \leq j \leq N$, so

$$\dim W_k^{2N} \geq (N+1) \dim W_{k-1}^N.$$

By induction, $\dim W_m^{2^m N} \geq (N+1)^m$, and $\operatorname{GKdim} E \geq m$. So $C(B)$ is a complete intersection of growth type with $\operatorname{gci}(C(B)) = m$. □

The bound $\operatorname{gci}(T) \leq \operatorname{gci}(T/(\Omega)) \leq \operatorname{gci}(T)+1$ for a regular normal element Ω [26, Theorem 1.11(a)], applied to $T = C^{(i-1)}[\bar{x}_i;\bar{\sigma}_i]$ and $\Omega = \bar{x}_i^{r_i}$, gives only inequalities at each step. The equality uses that $\bar{x}_i^{r_i}$ has degree at least two, through the periodicity class η. In the same way, Proposition 6.4 shows that the Ext-algebra of $C(B)$ has the same bigraded Hilbert series as the Ext-algebra of the commutative complete intersection $\bigotimes_{c_i \neq e} \Bbbk[x_i]/(x_i^{r_i})$, which is the covariant ring of the diagonal reflection group $\prod_i \mu_{r_i}$.

Example 6.7. In Example 2.21, the grading of $\Bbbk[x_1,\ldots,x_n]$ has homogeneous variables, and $C(B) = \bigotimes_i \Bbbk[x_i]/(x_i^{r_i})$. For $\Gamma = \Bbbk[x]/(x^r)$ with $r \geq 2$, the resolution of Proposition 6.4 with $C' = \Bbbk$ is the periodic resolution

$$\cdots \to \Gamma(-2r) \xrightarrow{x^{r-1}} \Gamma(-r-1) \xrightarrow{x} \Gamma(-r) \xrightarrow{x^{r-1}} \Gamma(-1) \xrightarrow{x} \Gamma \to \Bbbk \to 0,$$

and $E(\Gamma)$ is generated by the class $\xi \in \operatorname{Ext}^1$ and the periodicity class $\eta \in \operatorname{Ext}^2$. Theorem D gives $\operatorname{gci}(C(B)) = \#\{i \mid r_i \geq 2\}$, which agrees with the Künneth formula.

Example 6.8. Let $B = \mathcal{O}_q(\Bbbk^n)$ with a grading as in Corollary 5.9, so that $G = \prod_i \langle c_i \rangle$ and B_e is generated by $x_1^{r_1}, \dots, x_n^{r_n}$. Then $B(B_e)_{\geq 1} = \sum_i B x_i^{r_i}$, and

$$C(B) = \mathcal{O}_q(\Bbbk^n)/\big(x_1^{r_1}, \dots, x_n^{r_n}\big), \quad H_{C(B)}(s) = \prod_{i=1}^{n} [r_i]_s,$$

a quantum complete intersection. Each $x_i^{r_i}$ is normal, since $x_i^{r_i} x_j = q_{ij}^{r_i} x_j x_i^{r_i}$. For $0 \leq k < n$, the quotient $\mathcal{O}_q(\Bbbk^n)/(x_1^{r_1}, \dots, x_k^{r_k})$ has the basis of monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_i < r_i$ for $i \leq k$, and multiplication by $x_{k+1}^{r_{k+1}}$ sends such a monomial to a nonzero multiple of another one. So $x_1^{r_1}, \dots, x_n^{r_n}$ is a regular sequence of normal elements, and $C(B)$ is a classical complete intersection in the sense of [26, Definition 1.3]. The Ext-algebras of these quantum complete intersections were determined by Bergh and Oppermann [7, Theorem 5.3]. By Theorem D,

$$P_{C(B)}(y, s) = \prod_{r_i \geq 2} \frac{1 + ys}{1 - y^2 s^{r_i}}, \quad \operatorname{gci}\big(C(B)\big) = \#\{i \mid r_i \geq 2\}.$$

Example 6.9. For the algebra $B = \Bbbk[x][y; \sigma_2][z; \sigma_3, \delta_3]$ of Example 4.11 with its $\mathbb{Z}_2$-grading, we have $c_1 = c_3 = e$ and $c_2 = \tau$. Theorem D gives $C(B) \cong \Bbbk[\bar{y}]/(\bar{y}^2)$, $P_{C(B)}(y, s) = (1 - ys)^{-1}$ and $\operatorname{gci}(C(B)) = 1$. The step with $\delta_3 \neq 0$ contributes nothing.

Example 6.10. Let B be the algebra (5.12) with $q = \pm 1$, graded by $G = (\mathbb{Z}_r \times \mathbb{Z}_r) \rtimes \mathbb{Z}_{2m}$ as in Example 5.11, and write $a = \bar{x}_1$, $b = \bar{x}_2$ and $c = \bar{x}_3$. By Theorem D, $C(B)$ is presented by

$$a^r = b^r = c^{2m} = 0, \quad ba = q\, ab, \quad ca = \lambda\, bc, \quad cb = \mu\, ac,$$

it has the basis $a^i b^j c^k$ with $0 \leq i, j < r$ and $0 \leq k < 2m$, and

$$H_{C(B)}(s) = [r]_s^2\, [2m]_s, \quad \operatorname{gci}\big(C(B)\big) = 3,$$
$$P_{C(B)}(y, s) = \Big(\frac{1 + ys}{1 - y^2 s^r}\Big)^2 \frac{1 + ys}{1 - y^2 s^{2m}}.$$

The algebra $C(B)$ is quadratic exactly when $r = 2$ and $m = 1$. In that case, for the degree-lexicographic order with $c > b > a$, the leading words of the six relations are aa, bb, cc, ba, ca, cb, the words avoiding them are the eight words $a^i b^j c^k$ with $i, j, k \in \{0, 1\}$, and $\dim C(B) = |G| = 8$. So the relations form a quadratic Gröbner basis, $C(B)$ is Koszul [38, Theorem 5.3], and for the pairing $\langle \alpha\beta, ab \rangle = 1$ and the convention $E(C) \cong C^!$ for the Yoneda product its Ext-algebra is

$$E\big(C(B)\big) = \Bbbk\langle \alpha, \beta, \kappa \rangle / \big(\alpha\beta + q\beta\alpha,\ \kappa\alpha + \lambda^{-1}\beta\kappa,\ \kappa\beta + \mu^{-1}\alpha\kappa\big),$$

with Hilbert series $(1 - s)^{-3}$. For $r > 2$ or $m > 1$, the algebra $C(B)$ is not quadratic, and Theorem D still gives its Poincaré series.

Example 6.11. For the family of Example 5.13, Theorem D gives

$$C(B) \cong \Big(\bigotimes_{i=1}^{n} \Bbbk[\bar{x}_i]/(\bar{x}_i^{\,r})\Big)[\bar{t}; \bar{\sigma}]/(\bar{t}^{\,np}), \quad \bar{\sigma}(\bar{x}_i) = \bar{x}_{i+1},$$

with Hilbert series $[r]_s^n [np]_s$, and for $r \geq 2$,

$$P_{C(B)}(y, s) = \Big(\frac{1 + ys}{1 - y^2 s^r}\Big)^n \frac{1 + ys}{1 - y^2 s^{np}}, \quad \operatorname{gci}\big(C(B)\big) = n + 1.$$

We make the case $n = r = 2$, $p = 1$ of Example 5.17 explicit. Write $a = \bar{x}_1$ and $b = \bar{x}_2$. Then $C = C(B)$ is presented by

$$a^2 = b^2 = t^2 = 0, \quad ba = ab, \quad ta = bt, \quad tb = at.$$

For the degree-lexicographic order with $t > b > a$, the leading words of these six relations are aa, bb, tt, ba, ta, tb. The words avoiding them are those with strictly increasing letters, that is, $a^i b^j t^k$ with $i, j, k \in \{0, 1\}$. Reducing with the six relations shows that these eight words span C. Since $\dim C = |G| = 8$, they are linearly independent, so no further leading word occurs in the ideal of relations. Hence the six relations form a quadratic Gröbner basis, C has a PBW basis, and C is Koszul [38, Theorem 5.3]. Its Ext-algebra is the quadratic dual

$$E(C) = C^! = \Bbbk\langle \alpha, \beta, \tau \rangle / (\alpha\beta + \beta\alpha,\ \tau\alpha + \beta\tau,\ \tau\beta + \alpha\tau),$$

since the orthogonal complement of the span of $aa, bb, tt, ba-ab, ta-bt, tb-at$ is spanned by $\alpha\beta+\beta\alpha$, $\tau\alpha+\beta\tau$ and $\tau\beta+\alpha\tau$. The algebra $C^!$ is the Ore extension $\Bbbk_{-1}[\alpha,\beta][\tau;\vartheta]$ with $\vartheta(\alpha)=-\beta$ and $\vartheta(\beta)=-\alpha$, which is AS regular of dimension three. Hence

$$H_C(s)=(1+s)^3, \quad H_{E(C)}(s)=\frac{1}{H_C(-s)}=\frac{1}{(1-s)^3}, \quad \dim \operatorname{Ext}^q_C(\Bbbk,\Bbbk)=\binom{q+2}{2},$$

in agreement with $P_{C(B)}(y,s)=(1+ys)^3/(1-y^2s^2)^3=(1-ys)^{-3}$. In low degrees,

q	0	1	2	3	4	5
$\dim \operatorname{Ext}^q_C(\Bbbk,\Bbbk)$	1	3	6	10	15	21

and $\operatorname{gci}(C)=3$ for this nonabelian group. In this case C is also a classical complete intersection. The ideal $B(B_e)_{\geq 1}$ is generated by x_1^2, x_2^2 and t^2, hence by

$$\Omega_1=t^2, \quad \Omega_2=x_1^2+x_2^2, \quad \Omega_3=x_1^2-x_2^2.$$

Here Ω_1 and Ω_2 are central in B, since $tx_1^2=x_2^2t$ and $tx_2^2=x_1^2t$, and $t\Omega_3=-\Omega_3 t$, so Ω_3 is normal. As left A-modules, $B/(\Omega_1)=A\oplus At$ with $A=\Bbbk[x_1,x_2]$, and $\Omega_2,\Omega_3\in A$ act on both summands by left multiplication. Since Ω_2,Ω_3 is a regular sequence in A, the sequence $\Omega_1,\Omega_2,\Omega_3$ of normal elements is regular in B, and $C=B/(\Omega_1,\Omega_2,\Omega_3)$.

The following table records the identity components, the covariant rings and the values of $\operatorname{gci}(C(B))$ in the examples above. In each case $\operatorname{gci}(C(B))$ is the number of generators of nontrivial G-degree.

B	G	B_e	$C(B)$	$\operatorname{gci}(C(B))$
$\Bbbk[x_1,\dots,x_n]$	$\prod_i \mathbb{Z}_{r_i}$	$\Bbbk[x_1^{r_1},\dots,x_n^{r_n}]$	$\bigotimes_i \Bbbk[\bar{x}_i]/(\bar{x}_i^{\,r_i})$	$\#\{i \mid r_i\geq 2\}$
$\mathcal{O}_q(\Bbbk^n)$	$\prod_i \mathbb{Z}_{r_i}$	$\mathcal{O}_{q'}(\Bbbk^n)$	$\mathcal{O}_q(\Bbbk^n)/(x_i^{r_i})$	$\#\{i \mid r_i\geq 2\}$
Jordan plane	$\{e\}$	B	$\Bbbk$	0
Example 4.11	$\mathbb{Z}_2$	$\Bbbk[x,w][z;\delta_e]$	$\Bbbk[\bar{y}]/(\bar{y}^{\,2})$	1
Example 4.14	$\mathbb{Z}_2$	$\Bbbk[x,w][z;\sigma,\delta]$	$\Bbbk[\bar{y}]/(\bar{y}^{\,2})$	1
Example 5.11	$(\mathbb{Z}_r\times\mathbb{Z}_r)\rtimes\mathbb{Z}_{2m}$	skew polynomial	Example 6.10	3
$\Bbbk[x_1,\dots,x_n][t;\sigma]$	$\mathbb{Z}_r^n\rtimes\mathbb{Z}_{np}$, $r\geq 2$	$\Bbbk[x_1^r,\dots,x_n^r,t^{np}]$	Example 6.11	$n+1$

7. McKay quivers

In this section we describe the smash products $B\#\Bbbk^G$ and $C(B)\#\Bbbk^G$ by quivers with relations, we compute their matrix Hilbert series, and we invert the Jacobian. This proves Theorem E.

Let G be a finite group and B a G-graded algebra. Let $\{p_g\}_{g\in G}$ be the basis of $\Bbbk^G$ dual to G, so that $p_gp_h=\delta_{g,h}p_g$. The smash product $\Lambda=B\#\Bbbk^G$ is the vector space $B\otimes\Bbbk^G$ with the multiplication

$$(a\#p_g)(b\#p_h)=a\,b_{gh^{-1}}\#p_h \quad (a,b\in B,\ g,h\in G), \tag{7.1}$$

where $b_{gh^{-1}}$ is the component of b of degree gh^{-1} [14, Proposition 1.4]. Both ways of computing the product of $a\#p_g$, $b\#p_h$ and $c\#p_k$ give the element $a\,b_{gh^{-1}}c_{hk^{-1}}\#p_k$, and the unit is $\sum_g 1\#p_g$. The category of graded right B-modules is isomorphic to the category of right Λ-modules [14, Theorem 2.2]. Put

$$e_g=1\#p_g \quad (g\in G).$$

These are orthogonal idempotents with $\sum_g e_g=1$, and (7.1) gives

$$e_g\Lambda e_h=B_{gh^{-1}}\#p_h\cong B_{gh^{-1}} \quad (g,h\in G). \tag{7.2}$$

For $a,b\in B_e$ we have $(a\#p_h)(b\#p_h)=ab\#p_h$, so every corner $e_h\Lambda e_h$ is isomorphic to B_e as an algebra. In particular, if B is a noetherian AS regular domain generated in degree one with a faithful G-grading refining the $\mathbb{N}$-grading, then G is a dual reflection group for B if and only if one, or equivalently every, corner $e_h\Lambda e_h$ is AS regular. If B is $\mathbb{N}$-graded and the G-grading refines the $\mathbb{N}$-grading, then Λ is $\mathbb{N}$-graded by $\deg(b\#p_h)=\deg b$, and its *matrix Hilbert series* is

$$H_\Lambda(s)=\big(H_{e_g\Lambda e_h}(s)\big)_{g,h\in G}=\big(H_{B_{gh^{-1}}}(s)\big)_{g,h\in G}. \tag{7.3}$$

If G is abelian, then $\Bbbk^G\cong\Bbbk G^\vee$ as Hopf algebras, and Λ is the skew group algebra $B\#\Bbbk G^\vee$ for the action of the dual group described in Section 2.

Suppose that B is generated by B_1, and let $x_1, \dots, x_n$ be a basis of B_1 consisting of G-homogeneous elements, with x_i of degree c_i. Such a basis exists when the G-grading refines the $\mathbb{N}$-grading. Let Q_G be the quiver with vertex set G and with arrows

$$x_i^{(h)} \colon h \longrightarrow c_i h \quad (h \in G,\ 1 \le i \le n).$$

This is the McKay quiver of the $\Bbbk^G$-module B_1 in the sense of [15], and it is the Cayley quiver of G with respect to $c_1, \dots, c_n$. It is connected when the c_i generate G. We compose paths from right to left, and we send the trivial path at g to e_g and the arrow $x_i^{(h)}$ to $x_i \# p_h \in e_{c_i h} \Lambda e_h$. By (7.1),

$$(x_{i_k} \# p_{h_k}) \cdots (x_{i_1} \# p_{h_1}) = x_{i_k} \cdots x_{i_1} \# p_{h_1} \quad \text{if } h_{j+1} = c_{i_j} h_j \text{ for all } j, \tag{7.4}$$

and the product is zero otherwise. For a G-homogeneous element ρ of the tensor algebra $T(B_1)$ and $h \in G$, we write $\rho^{(h)}$ for the linear combination of paths starting at h which corresponds to $\rho \# p_h$ under (7.4).

Let $B = T(V)/I$ with $V = B_1$, where I is generated by elements $\rho_1, \dots, \rho_k$ that are homogeneous for the $\mathbb{N}$-grading and for the G-grading. Then the assignment above induces an isomorphism of graded algebras

$$\Lambda = B \# \Bbbk^G \cong \Bbbk Q_G / I_G, \tag{7.5}$$

where I_G is the ideal generated by the elements $\rho_j^{(h)}$ for $1 \le j \le k$ and $h \in G$. This is the case of a quiver with one vertex of the presentation of smash products of group weighted bound quivers by Asashiba [6, Theorem 1.18], the smash product of a G-graded algebra with G being identified with $B \# \Bbbk^G$ by [13, Proposition 3.7]. It is also an instance of the theory of coverings of quivers with relations [20], and of Crawford's description of smash products of Koszul AS regular algebras with semisimple Hopf algebras by quivers with relations [15, Theorem 1.2]. For iterated Ore extensions it takes the following explicit form.

Corollary 7.6. *Let B be the iterated Ore extension* (1.5) *with a G-grading in which every x_i is homogeneous. Then $\Lambda = B \# \Bbbk^G$ is the path algebra of the Cayley quiver Q_G of G with respect to $c_1, \dots, c_n$, modulo the ideal generated by*

$$x_i^{(c_j h)} x_j^{(h)} - \sigma_i(x_j)^{(c_i h)} x_i^{(h)} - \delta_i(x_j)^{(h)} \quad (1 \le j < i \le n,\ h \in G), \tag{7.7}$$

where $\sigma_i(x_j) \in B_1$ and $\delta_i(x_j)$ is replaced by a G-homogeneous lift in $V \otimes V$ of degree $c_i c_j$. All three terms of (7.7) *are paths from h to $c_i c_j h$. If moreover G is a dual reflection group for B and $(B_e)_1 = 0$, then the relations are the binomial squares*

$$x_i^{(c_j h)} x_j^{(h)} = \lambda_{ij}\, x_{\pi_i(j)}^{(c_i h)} x_i^{(h)} \quad (1 \le j < i \le n,\ h \in G),$$

with λ_{ij} and π_i as in Theorem C.

Proof. The relations (2.11) are homogeneous for both gradings, since $\sigma_i(x_j)$ has degree $c_i c_j c_i^{-1}$ and $\delta_i(x_j)$ has degree $c_i c_j$ by (2.13). Since $T(V) \to B$ is a surjection of G-graded spaces, homogeneous lifts exist, and (7.5) applies. The term $\sigma_i(x_j)^{(c_i h)} x_i^{(h)}$ ends at $c_i c_j c_i^{-1} \cdot c_i h = c_i c_j h$. The second statement follows from the binomial presentation of Theorem C, since $c_{\pi_i(j)} c_i = c_i c_j$. □

We compute the matrix Hilbert series. For $c \in G$, let Π_c be the permutation matrix of left multiplication by c, with rows and columns indexed by G,

$$(\Pi_c)_{g,h} = \begin{cases} 1 & g = ch, \\ 0 & \text{otherwise}, \end{cases} \qquad \Pi_c \Pi_{c'} = \Pi_{cc'}, \quad \Pi_c^{|c|} = I. \tag{7.8}$$

Left multiplication by c has $|G|/|c|$ orbits on G, the right cosets of $\langle c \rangle$, each of length $|c|$. Hence

$$\det(I - s\Pi_c) = (1 - s^{|c|})^{|G|/|c|}, \quad (I - s\Pi_c)^{-1} = \frac{1}{1 - s^{|c|}} \sum_{k=0}^{|c|-1} s^k \Pi_c^k. \tag{7.9}$$

Proposition 7.10. *Let B be the iterated Ore extension* (1.5) *with a G-grading in which every x_i is homogeneous, and let $\Lambda = B \# \Bbbk^G$. Then*

$$H_\Lambda(s) = (I - s\Pi_{c_1})^{-1} (I - s\Pi_{c_2})^{-1} \cdots (I - s\Pi_{c_n})^{-1}, \tag{7.11}$$

so that $H_\Lambda(s)^{-1} = (I - s\Pi_{c_n}) \cdots (I - s\Pi_{c_1})$ and $\det H_\Lambda(s)^{-1} = \prod_{i=1}^n (1 - s^{r_i})^{|G|/r_i}$.

Proof. The monomials $x_1^{a_1}\cdots x_n^{a_n}$ with $a_i \geq 0$ form a basis of B, and such a monomial has $\mathbb{N}$-degree $a_1+\cdots+a_n$ and G-degree $c_1^{a_1}\cdots c_n^{a_n}$. By (7.3) and (7.8), the (g,h) entry of $H_\Lambda(s)$ is

$$\sum_{c_1^{a_1}\cdots c_n^{a_n}=gh^{-1}} s^{a_1+\cdots+a_n} = \sum_{a_1,\ldots,a_n\geq 0} s^{a_1+\cdots+a_n}\big(\Pi_{c_1}^{a_1}\cdots\Pi_{c_n}^{a_n}\big)_{g,h}.$$

Summing the geometric series in each a_i gives (7.11), and the determinant follows from (7.9). □

Thus $H_\Lambda(s)^{-1}$ is a matrix polynomial of degree n, whose factors $I - s\Pi_{c_i}$ involve the adjacency matrices Π_{c_i} of the arrows of each type, one for each Ore step. For a dual reflection group, the matrix Hilbert series also factors through the identity component.

Proposition 7.12. *In the setting of Proposition 7.10, suppose that G is a dual reflection group for B, and put $P_G(s) = \big(s^{\deg f_{gh^{-1}}}\big)_{g,h\in G}$. Then*

$$\begin{aligned} H_\Lambda(s) &= H_{B_e}(s)\, P_G(s), \\ P_G(s) &= \prod_{i=1}^{n}\Big(\sum_{k=0}^{r_i-1} s^k \Pi_{c_i}^k\Big), \\ \det P_G(s) &= \prod_{i=1}^{n}(1-s^{r_i})^{|G|(r_i-1)/r_i}. \end{aligned} \tag{7.13}$$

Proof. By (2.18), $H_{B_{gh^{-1}}}(s) = s^{\deg f_{gh^{-1}}} H_{B_e}(s)$, which gives the first identity by (7.3). By (7.9) and Proposition 7.10,

$$H_\Lambda(s) = \prod_{i=1}^{n}\frac{1}{1-s^{r_i}}\cdot\prod_{i=1}^{n}\Big(\sum_{k=0}^{r_i-1} s^k\Pi_{c_i}^k\Big),$$

where the scalar factors have been moved to the front. The first factor is $H_{B_e}(s)$ by Theorem C, and it is an invertible power series, which gives the factorization of P_G. The determinant follows from (7.9), since $\det\big(\sum_{k<r} s^k\Pi_c^k\big) = (1-s^r)^{|G|}\det(I-s\Pi_c)^{-1} = (1-s^r)^{|G|-|G|/r}$ for $r=|c|$. □

The factorization of P_G also follows directly from the normal form of Theorem C(3). The (g,h) entry of the product is $\sum s^{a_1+\cdots+a_n}$ over the tuples with $0\leq a_i<r_i$ and $c_1^{a_1}\cdots c_n^{a_n} = gh^{-1}$, and by uniqueness of the normal form there is exactly one such tuple, with $a_1+\cdots+a_n = \deg f_{gh^{-1}}$ by (5.2). Each column of $P_G(s)$ sums to $p_G(s) = \prod_i [r_i]_s$, so (7.13) is the matrix form of the factorization of p_G.

Let G be a dual reflection group for B. The ideal $J = B(B_e)_{\geq 1}$ is two-sided by (2.16), and it is G-graded. So $J\#\Bbbk^G$ is an ideal of Λ, and

$$\Lambda/(J\#\Bbbk^G) \cong C(B)\#\Bbbk^G.$$

Proposition 7.14. *Let B be a noetherian AS regular domain generated in degree one, G a dual reflection group for B, and $\bar\Lambda = C(B)\#\Bbbk^G$.*

1. *The algebra $\bar\Lambda$ has dimension $|G|^2$, and for all $g,h\in G$, the space $e_g\bar\Lambda e_h = \Bbbk\,\bar f_{gh^{-1}}\#p_h$ is one-dimensional, concentrated in degree $\deg f_{gh^{-1}}$. The matrix Hilbert series of $\bar\Lambda$ is $P_G(s)$.*
2. *The algebra $\bar\Lambda$ is Frobenius. The socle of the left projective module $\bar\Lambda e_h$ is $e_{m_G h}\bar\Lambda e_h$, which is isomorphic to the simple module at the vertex $m_G h$.*

Proof. (1) By (2.16), $C(B)_{gh^{-1}} = \Bbbk\bar f_{gh^{-1}}$, and (7.2) for $C(B)$ gives the claim.

(2) Since $C(B)$ is Frobenius by (2.16), so is $\bar\Lambda$ by [18, Corollary 2.2]. The projective module $\bar\Lambda e_h = \bigoplus_g e_g\bar\Lambda e_h$ is indecomposable since $e_h\bar\Lambda e_h = \Bbbk e_h$. Its component $e_{m_G h}\bar\Lambda e_h = \Bbbk\,\bar f_{m_G}\#p_h$ has the top degree N_G. Since $\bar\Lambda$ is finite-dimensional and $\bar\Lambda_0 = \bigoplus_g \Bbbk e_g$ is semisimple, the radical of $\bar\Lambda$ is $\bar\Lambda_{\geq 1}$. So $e_{m_G h}\bar\Lambda e_h$ is annihilated by the radical and lies in the socle. The socle of an indecomposable projective module over a Frobenius algebra is simple, so it equals $e_{m_G h}\bar\Lambda e_h$. □

For the iterated Ore extensions of Theorem C, the algebra B_e is generated by the elements $x_i^{r_i}$ by Proposition 5.6, so J is the two-sided ideal generated by $x_1^{r_1},\ldots,x_n^{r_n}$. Then Corollary 7.6 shows that $\bar\Lambda$ is the quotient of $\Bbbk Q_G/I_G$ by the paths

$$x_i^{(c_i^{r_i-1}h)}\cdots x_i^{(c_ih)}x_i^{(h)} \quad (1\leq i\leq n,\ h\in G)$$

of length r_i, where $r_i = 1$ when $\delta_i \neq 0$.

We invert the Jacobian. Let G be a dual reflection group for B. By (2.16), f_{m_G} is normal, so there is a graded automorphism ν of B with $f_{m_G} b = \nu(b) f_{m_G}$, and comparing G-degrees in the domain B gives $\nu(B_g) = B_{m_G g m_G^{-1}}$. The set $\{f_{m_G}^k\}_{k\geq 0}$ is a two-sided Ore set, and the localization

$$L = B[f_{m_G}^{-1}]$$

is a G-graded domain containing B, with $f_{m_G}^{-1}$ homogeneous of degree m_G^{-1}.

Proposition 7.15. *Let B be a noetherian AS regular domain generated in degree one, G a dual reflection group for B, and $L = B[f_{m_G}^{-1}]$. Let o be the order of m_G and $F = f_{m_G}^o$.*

1. *Every f_g is a unit of L, and $L_g = f_g L_e = L_e f_g$ for all $g \in G$. So L is a crossed product $L_e * G$, and $L_g L_h = L_{gh}$ for all $g, h \in G$.*
2. *The element F is a normal element of B_e, and $L_e = B_e[F^{-1}]$.*
3. *The elements*

$$u_{g,h} = f_g f_h^{-1} \# p_h \in e_g(L \# \Bbbk^G) e_h \quad (g, h \in G)$$

form a complete set of matrix units of $L\#\Bbbk^G$, and $L\#\Bbbk^G \cong M_{|G|}(L_e)$.

Proof. (1) By (2.17), the element f_g has the right inverse $\mu_g^{-1} f_{g^{-1} m_G} f_{m_G}^{-1}$ in L, which is homogeneous of degree g^{-1}. In a domain a one-sided inverse is two-sided, since $ab = 1$ implies $a(ba - 1) = 0$. So f_g is a unit with $f_g^{-1} \in L_{g^{-1}}$, and for $b \in L_g$ we have $b = f_g(f_g^{-1} b)$ with $f_g^{-1} b \in L_e$, and similarly on the other side. Then $L_{gh} = f_g f_h L_e \subseteq L_g L_h \subseteq L_{gh}$.

(2) The element F has degree $m_G^o = e$, so $F \in B_e$, and $F B_e = \nu^o(B_e) F = B_e F$, since ν preserves B_e. An element of L_e has the form $b f_{m_G}^{-k}$ with $b \in B$, and comparing degrees gives $b \in B_{m_G^k}$. Choose l with $lo \geq k$. Then $b f_{m_G}^{-k} = \left(b f_{m_G}^{lo-k}\right) F^{-l}$ with $b f_{m_G}^{lo-k} \in B_{m_G^{lo}} = B_e$. Hence $L_e = B_e[F^{-1}]$.

(3) For $g, h, h', k \in G$, the element $f_{h'} f_k^{-1}$ is homogeneous of degree $h'k^{-1}$, so (7.1) gives

$$u_{g,h} u_{h',k} = f_g f_h^{-1} \left(f_{h'} f_k^{-1}\right)_{hk^{-1}} \# p_k = \begin{cases} f_g f_k^{-1} \# p_k = u_{g,k} & h = h', \\ 0 & h \neq h'. \end{cases}$$

Moreover $\sum_g u_{g,g} = \sum_g 1 \# p_g = 1$. A ring with a complete set of $N \times N$ matrix units is isomorphic to the matrix ring of size N over the corner of $u_{e,e}$. Here $u_{e,e} = e_e$ since $f_e = 1$, and $e_e(L\#\Bbbk^G)e_e \cong L_e$ by (7.2) for L. □

The identity (2.17) is due to Kirkman and Zhang [23, Lemma 2.11(2)]. The Morita equivalence between $L\#\Bbbk^G$ and L_e also follows from the fact that L is strongly graded [14, Theorem 2.12], and the matrix units of Proposition 7.15(3) make it an explicit isomorphism. In Example 2.21, $L = \Bbbk[x_i^{\pm 1}\ (r_i > 1),\ x_j\ (r_j = 1)]$, the algebra Λ is the skew group algebra of the diagonal reflection group $\prod_i \mu_{r_i}$, and Proposition 7.15 recovers the Morita equivalence between $L\#\Bbbk G^\vee$ and $L^{G^\vee} = L_e$ on the complement of the reflecting hyperplanes $x_i = 0$ with $r_i > 1$, on which the group acts freely [44, Theorem 1.5].

Proof of Theorem E. Part (1) is Corollary 7.6 and Proposition 7.10. Part (2) is Propositions 7.12 and 7.14, and part (3) is Proposition 7.15. □

Example 7.16. For $\Bbbk[x]$ graded by $\mathbb{Z}_r = \langle c \rangle$ with x of degree c, the quiver Q_G is the cyclic quiver with r vertices and no relations, so $\Bbbk[x]\#\Bbbk^{\mathbb{Z}_r}$ is its path algebra (Figure 1), and (7.11) reads $H_\Lambda(s)^{-1} = I - s\Pi_c$, where Π_c is the adjacency matrix of the cyclic quiver. The matrix $P_G(s)$ is the circulant matrix with first column $(1, s, \ldots, s^{r-1})$, and (7.13) reduces to the classical formula $\det P_G(s) = (1 - s^r)^{r-1}$. The algebra $\bar{\Lambda}$ is the path algebra of the cyclic quiver modulo all paths of length r, a self-injective Nakayama algebra, and the socle of its projective at c^j sits at c^{j+r-1}, in agreement with $m_G = c^{r-1}$. Moreover $L = \Bbbk[x^{\pm 1}]$ and $L_e = \Bbbk[x^{\pm r}]$ for $r \geq 2$.

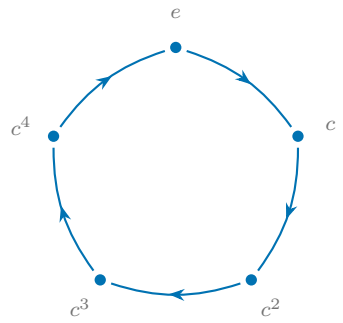


FIGURE 1. The McKay quiver of $\Bbbk[x]$ graded by $\mathbb{Z}_5$.

Example 7.17. For the quantum plane $\Bbbk_q[x,y]$ with $yx = qxy$, graded by $\mathbb{Z}_{r_1} \times \mathbb{Z}_{r_2}$ with $\deg x = (1,0)$ and $\deg y = (0,1)$, the quiver Q_G is the toroidal grid with vertices (a,b) and arrows

$$x^{(a,b)}\colon (a,b) \to (a+1,b), \quad y^{(a,b)}\colon (a,b) \to (a,b+1),$$

and I_G is generated by the q-commutative squares

$$y^{(a+1,b)}x^{(a,b)} = q\,x^{(a,b+1)}y^{(a,b)} \quad \big((a,b) \in \mathbb{Z}_{r_1} \times \mathbb{Z}_{r_2}\big),$$

see Figure 2. Crawford describes the same toroidal pictures for coactions on two-dimensional AS regular algebras [16, Section 4], using the presentation of [16, Theorem 2.6], with arrows $g \to a^{-1}g$ and paths composed from left to right. For $g = (a,b)$ we have $f_g = x^a y^b$ and $g^{-1}m_G = (r_1 - 1 - a, r_2 - 1 - b)$. Since $y^b x^c = q^{bc}x^c y^b$,

$$f_g f_{g^{-1}m_G} = x^a y^b\, x^{r_1-1-a} y^{r_2-1-b} = q^{b(r_1-1-a)}\, x^{r_1-1} y^{r_2-1},$$

so $\mu_g = q^{b(r_1-1-a)}$ in (2.17). If $r_1, r_2 \geq 2$, then L is the quantum torus $\Bbbk_q[x^{\pm 1}, y^{\pm 1}]$, with identity component $L_e = \Bbbk_{q^{r_1 r_2}}[x^{\pm r_1}, y^{\pm r_2}]$.

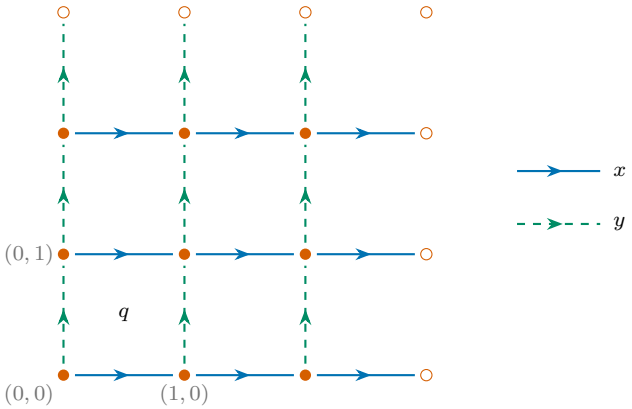


FIGURE 2. The McKay quiver of $\Bbbk_q[x,y]$ graded by $\mathbb{Z}_3 \times \mathbb{Z}_3$. The white vertices are identified with the vertices on the opposite side.

Example 7.18. We describe the McKay quiver of the $\mathbb{Z}_2$-grading of Example 4.11, where $\delta_3 \neq 0$. Here $B = \Bbbk[x][y;\sigma_2][z;\sigma_3,\delta_3]$ with relations (4.12), $\mathbb{Z}_2 = \{e,\tau\}$, $c_1 = c_3 = e$ and $c_2 = \tau$. The quiver Q_G has the two vertices e and τ, a loop $x^{(h)}$ and a loop $z^{(h)}$ at each vertex h, and arrows $y^{(e)}\colon e \to \tau$ and $y^{(\tau)}\colon \tau \to e$. By Corollary 7.6, the ideal I_G is generated by the six relations

$$y^{(h)}x^{(h)} = -x^{(\tau h)}y^{(h)}, \quad z^{(\tau h)}y^{(h)} = -y^{(h)}z^{(h)} - 2x^{(\tau h)}y^{(h)},$$
$$z^{(h)}x^{(h)} = x^{(h)}z^{(h)} + x^{(h)}x^{(h)} - y^{(\tau h)}y^{(h)},$$

for $h \in \{e, \tau\}$. The last family contains the lift of $\delta_3(x) = x^2 - y^2$, whose two terms are paths from h to h. By Propositions 7.10 and 7.12,

$$H_\Lambda(s) = \frac{1}{(1-s)^2}(I - s\Pi_\tau)^{-1} = \frac{1}{(1-s)^2(1-s^2)}\begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}, \quad P_G(s) = I + s\Pi_\tau,$$

which agrees with $H_{B_e}(s) = (1-s)^{-2}(1-s^2)^{-1}$. The loops $x^{(h)}$ and $z^{(h)}$ correspond to generators of trivial degree, so they are killed in $\bar{\Lambda}$, and $\bar{\Lambda} = C(B)\#\Bbbk^{\mathbb{Z}_2}$ is the path algebra of the cyclic quiver with two vertices modulo the paths of length two. The Jacobian $f_{m_G} = y$ has degree τ of order two, $F = y^2 = w$, and Proposition 7.15 gives

$$L_e = B_e[w^{-1}] = \Bbbk[x, w^{\pm 1}][z; \delta_e], \quad L\#\Bbbk^{\mathbb{Z}_2} \cong M_2(L_e),$$

with δ_e as in (4.13).

Example 7.19. For the dihedral group of Example 5.17, the quiver Q_G has eight vertices and three arrows $x_1^{(h)}\colon h \to \varepsilon_1 h$, $x_2^{(h)}\colon h \to \varepsilon_2 h$ and $t^{(h)}\colon h \to \gamma h$ at every vertex. By Corollary 7.6 and (5.18), the relations are the twenty-four commutative squares

$$x_2^{(\varepsilon_1 h)} x_1^{(h)} = x_1^{(\varepsilon_2 h)} x_2^{(h)},$$
$$t^{(\varepsilon_1 h)} x_1^{(h)} = x_2^{(\gamma h)} t^{(h)}, \quad t^{(\varepsilon_2 h)} x_2^{(h)} = x_1^{(\gamma h)} t^{(h)} \quad (h \in G).$$

Here $\gamma\varepsilon_1 = \varepsilon_2\gamma$ and $\gamma\varepsilon_2 = \varepsilon_1\gamma$, so both sides of each square have the same endpoints. Since ε_1, ε_2 and γ are involutions, the arrows come in opposite pairs, and Q_G is a cube (Figure 3). The t-arrows join h and γh, so they join e to γ, ε_1 to $\varepsilon_2\gamma$, ε_2 to $\varepsilon_1\gamma$ and $\varepsilon_1\varepsilon_2$ to $\varepsilon_1\varepsilon_2\gamma$. On the bottom face $\langle\varepsilon_1\rangle \times \langle\varepsilon_2\rangle$, the x_1-arrows join e to ε_1, while on the top face they join γ to $\varepsilon_1\gamma$, which lies above ε_2. So the arrows of type x_1 and x_2 exchange their directions between the two faces. This twist is the conjugation by γ, and it is the reason why G is not abelian.

Order G as $e, \varepsilon_1, \varepsilon_2, \varepsilon_1\varepsilon_2, \gamma, \varepsilon_1\gamma, \varepsilon_2\gamma, \varepsilon_1\varepsilon_2\gamma$. The table of Example 5.17 gives

$$P_G(s) = \begin{pmatrix} 1 & s & s & s^2 & s & s^2 & s^2 & s^3 \\ s & 1 & s^2 & s & s^2 & s^3 & s & s^2 \\ s & s^2 & 1 & s & s^2 & s & s^3 & s^2 \\ s^2 & s & s & 1 & s^3 & s^2 & s^2 & s \\ s & s^2 & s^2 & s^3 & 1 & s & s & s^2 \\ s^2 & s^3 & s & s^2 & s & 1 & s^2 & s \\ s^2 & s & s^3 & s^2 & s & s^2 & 1 & s \\ s^3 & s^2 & s^2 & s & s^2 & s & s & 1 \end{pmatrix},$$

which factors as

$$P_G(s) = (I + s\Pi_{\varepsilon_1})(I + s\Pi_{\varepsilon_2})(I + s\Pi_\gamma),$$

with $\det P_G(s) = (1-s^2)^{12}$ and $H_\Lambda(s) = (1-s^2)^{-3}P_G(s)$. The algebra $\bar{\Lambda} = C(B)\#\Bbbk^G$ is $\Bbbk Q_G$ modulo the twenty-four squares and the twenty-four paths

$$x_1^{(\varepsilon_1 h)} x_1^{(h)}, \quad x_2^{(\varepsilon_2 h)} x_2^{(h)}, \quad t^{(\gamma h)} t^{(h)} \quad (h \in G).$$

There are 72 paths of length two, and $\dim \bar{\Lambda}_n = 8, 24, 24, 8$ for $n = 0, 1, 2, 3$, so $\dim \bar{\Lambda} = 64 = |G|^2$. Every indecomposable projective has Loewy layers of dimensions $1, 3, 3, 1$, and by Proposition 7.14 the socle of the projective at h is the simple module at $\varepsilon_1\varepsilon_2\gamma h$, the vertex opposite to h in the cube. The mass element $m_G = \varepsilon_1\varepsilon_2\gamma$ has order two, $F = (x_1x_2t)^2 = x_1^2x_2^2t^2$, and by Proposition 7.15

$$L_e = \Bbbk[x_1^2, x_2^2, t^2]\big[(x_1^2x_2^2t^2)^{-1}\big] = \Bbbk[x_1^{\pm 2}, x_2^{\pm 2}, t^{\pm 2}],$$
$$L\#\Bbbk^G \cong M_8(L_e).$$

So the localization $L\#\Bbbk^G$ of Λ is Morita equivalent to L_e, the coordinate ring of the complement of the three coordinate planes in $\operatorname{Spec} B_e$.

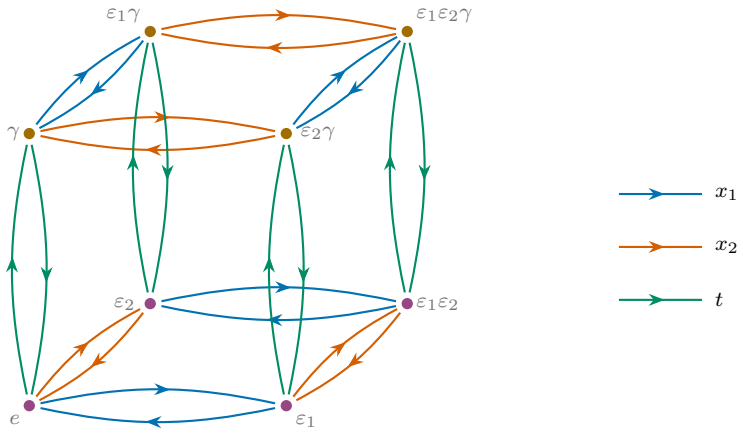


FIGURE 3. The McKay quiver of the dihedral group of order eight in Example 7.19.

8. Noncommutative quotients

In this section we interpret the extension $B_e \subseteq B$, the localization on which it becomes Hopf–Galois, and the point scheme of the dihedral example in the language of noncommutative algebraic geometry.

Artin and Zhang associate a noncommutative projective scheme to a noetherian connected graded algebra B through the quotient category $\operatorname{qgr} B$ [5]. We use the phrase *noncommutative affine space* only as a geometric analogy for an AS regular algebra B, and we write X/G informally for the object represented by the identity component B_e. A G-grading of B is an action of the Hopf algebra $\Bbbk^G$, whose invariant subalgebra is B_e. For G abelian it is an action of the character group $G^\vee$ by graded automorphisms (Section 2). In this algebraic sense Theorem C is a Chevalley–Shephard–Todd theorem for iterated Ore extensions. It characterizes when B_e is AS regular, and it identifies B_e with the iterated Ore extension of Proposition 5.6 on the generators $x_i^{r_i}$.

Let G be a dual reflection group for B. By (2.14), the extension $B_e \subseteq B$ is free of rank $|G|$ on both sides, with bases $\{f_g\}$. The augmentation of B_e gives

$$B \otimes_{B_e} \Bbbk = B/B(B_e)_{\geq 1} = C(B),$$

an algebra of dimension $|G|$, since $B(B_e)_{\geq 1}$ is a two-sided ideal by (2.16). It is the algebraic analogue of the fibre over the vertex of a commutative reflection quotient. For iterated Ore extensions, Theorem D shows that $C(B)$ is a complete intersection of growth type whose Ext-algebra has the same bigraded Hilbert series as the covariant ring of the diagonal reflection group $\prod_i \mu_{r_i}$, also when G is not abelian.

We identify a localization on which the extension becomes Hopf–Galois. A G-graded algebra R is a *$\Bbbk G$-Galois extension* of R_e if the canonical map

$$\beta\colon R \otimes_{R_e} R \longrightarrow R \otimes \Bbbk G, \quad a \otimes b \longmapsto \sum_{g \in G} a b_g \otimes g,$$

is bijective. This holds if and only if R is strongly graded, that is, $R_g R_h = R_{gh}$ for all $g, h \in G$ [35, Theorem 8.1.7]. It is a noncommutative analogue of the principal bundle condition. When G is abelian and R is commutative, it is the classical torsor condition for the action of the character group $G^\vee$. For nonabelian G it is understood in the Hopf–Galois sense.

Proposition 8.1. *Let B be a noetherian AS regular domain generated in degree one, G a dual reflection group for B, and $L = B[f_{m_G}^{-1}]$.*

1. *The algebra L is a $\Bbbk G$-Galois extension of L_e.*
2. *Suppose that B is the iterated Ore extension* (1.5) *with every x_i homogeneous. A ring homomorphism $\varphi\colon B \to R$ inverts f_{m_G} if and only if it inverts every x_i with $c_i \neq e$. In particular L is the localization of B at the generators of nontrivial G-degree, and $x_i^{-1} \in L_{c_i^{-1}}$ for every such i.*

Proof. (1) By Proposition 7.15(1), L is strongly graded, so L is a $\Bbbk G$-Galois extension of L_e by [35, Theorem 8.1.7].

(2) If $c_i \neq e$, then $r_i \geq 2$, and the normal form of c_i in Theorem C(3) is c_i itself, so $x_i = f_{c_i}$ by Theorem C. By Theorem C, $f_{m_G} = x_1^{r_1-1} \cdots x_n^{r_n-1}$, where only the factors with $c_i \neq e$ occur. If φ inverts these x_i, it inverts their product f_{m_G}. Conversely, if φ inverts f_{m_G}, then φ extends to the Ore localization L, and $\varphi(x_i) = \varphi(f_{c_i})$ is invertible since f_{c_i} is a unit of L by Proposition 7.15(1). The last statement follows since $x_i^{-1} = f_{c_i}^{-1} \in L_{c_i^{-1}}$. □

The generators x_i with $c_i \neq e$ need not be normal individually. In Example 5.17, for instance, $tx_1 = x_2 t$ does not lie in $x_1 B$, so we do not interpret their zero loci as noncommutative hypersurfaces. For the polynomial ring of Example 2.21, the grading is the diagonal action of $\prod_i \mu_{r_i}$, the algebra L is the coordinate ring of the complement of the reflecting hyperplanes $x_i = 0$ with $r_i > 1$, and the action is free on this complement [44, Theorem 1.5]. This agrees with the reflection arrangement $f_{m_G^{-1}}$ of Kirkman and Zhang [23, Definition 2.1], which in Examples 2.21 and 5.17 is the product of the generators of nontrivial G-degree.

Example 8.2. For the family of Example 5.13 with $r \geq 2$, every generator has nontrivial G-degree. By Proposition 8.1(2), L is the skew Laurent polynomial ring

$$L = \Bbbk[x_1^{\pm 1}, \ldots, x_n^{\pm 1}][t^{\pm 1}; \sigma], \quad L_e = \Bbbk[x_1^{\pm r}, \ldots, x_n^{\pm r}, t^{\pm np}].$$

We have $L_e = B_e[F^{-1}]$ by Proposition 7.15(2). The element $D = \varepsilon_1 \cdots \varepsilon_n$ is central of order r, and $m_G = D^{r-1}\gamma^{np-1} = D^{-1}\gamma^{-1}$ has order $o = \operatorname{lcm}(r, np)$. Since $x_1 \cdots x_n$ commutes with t,

$$F = f_{m_G}^{o} = (x_1 \cdots x_n)^{o(r-1)}\, t^{o(np-1)},$$

a monomial in $x_1^r, \ldots, x_n^r, t^{np}$ in which every variable occurs. Inverting F in the polynomial ring $B_e = \Bbbk[x_1^r, \ldots, x_n^r, t^{np}]$ therefore inverts each of these generators. Thus L_e is the coordinate ring of a commutative algebraic torus, while L is a noncommutative skew Laurent polynomial ring. The extension $L_e \subseteq L$ is Hopf–Galois for the coaction of $\Bbbk G$ with $G = \mathbb{Z}_r^n \rtimes \mathbb{Z}_{np}$, and $L\#\Bbbk^G \cong M_{npr^n}(L_e)$.

Let $B = T(V)/(\mathcal{R})$ be a quadratic algebra with $V = B_1$. A *point module* is a graded right B-module $M = \bigoplus_{j \geq 0} M_j$, generated by M_0, with $\dim M_j = 1$ for all j [4]. Choosing generators m_j of M_j, we have $m_j x = p_j(x) m_{j+1}$ for $x \in V$, where p_j is a nonzero linear form on V. Since the relations of B are quadratic, the sequence (p_j) of points of $\mathbb{P}(V^*)$ defines a point module if and only if

$$\sum_{a,b} \lambda_{ab}\, p_j(x_a)\, p_{j+1}(x_b) = 0 \quad \text{for every relation } \sum_{a,b} \lambda_{ab} x_a x_b \in \mathcal{R} \text{ and every } j \geq 0. \tag{8.3}$$

Example 8.4. Let $B = \Bbbk[x_1, x_2][t; \sigma]$ with $\sigma(x_1) = x_2$ and $\sigma(x_2) = x_1$, the algebra of the dihedral group in Example 5.17, with relations $x_2 x_1 = x_1 x_2$, $tx_1 = x_2 t$ and $tx_2 = x_1 t$. Write $p = (a_1 : a_2 : a_3)$ and $q = (b_1 : b_2 : b_3)$ for points of $\mathbb{P}(V^*) = \mathbb{P}^2$, with $a_1 = p(x_1)$, $a_2 = p(x_2)$ and $a_3 = p(t)$. By (8.3), consecutive points $p = p_j$ and $q = p_{j+1}$ of a point module satisfy $M(p)q = 0$, where

$$M(p) = \begin{pmatrix} a_2 & -a_1 & 0 \\ a_3 & 0 & -a_2 \\ 0 & a_3 & -a_1 \end{pmatrix}, \quad \det M(p) = a_3\,(a_2^2 - a_1^2).$$

So p_0 lies in the cubic

$$E = \{a_3 = 0\} \cup \{a_1 = a_2\} \cup \{a_1 = -a_2\} \subseteq \mathbb{P}^2,$$

a triangle of lines. The matrix $M(p)$ has rank exactly two at every point of E. If $a_3 \neq 0$, the minor formed by the second and third rows and the first and second columns equals a_3^2. If $a_3 = 0$, the first row is nonzero and is independent of at least one of the other two rows. So the kernel of $M(p)$ is one-dimensional on E, and q is uniquely determined by p. Since $\det M(p)$ is a product of three distinct linear forms, E is reduced. Solving $M(p)q = 0$ on each line gives an automorphism σ_E of E with $q = \sigma_E(p)$, namely

$$\sigma_E(a_1 : a_2 : 0) = (a_1 : a_2 : 0), \quad \sigma_E(a : a : c) = (a : a : c), \quad \sigma_E(a : -a : c) = (a : -a : -c).$$

On the line $a_1 = -a_2$, for instance, the equations read $-ab_1 - ab_2 = 0$, $cb_1 + ab_3 = 0$ and $cb_2 - ab_3 = 0$, whose solution is $q = (a : -a : -c)$ for $a \neq 0$, and $q = (0 : 0 : 1)$ for $a = 0$. The two formulas agree at the vertices $(1 : -1 : 0)$ and $(0 : 0 : 1)$, which are fixed. Hence the point modules of B are parametrized by E, and the point module of $p \in E$ is given by $p_j = \sigma_E^j(p)$.

The three components of E are the zero loci of the normal linear elements t, $x_1 - x_2$ and $x_1 + x_2$, since $t(x_1 \pm x_2) = \pm(x_1 \pm x_2)t$. On the component $a_1 = -a_2$, the restriction of σ_E coincides with the projective

action of the automorphism $t \mapsto -t$, $x_i \mapsto x_i$, attached to the character of G that is trivial on $\varepsilon_1, \varepsilon_2$ and takes the value -1 on γ. This automorphism does not induce σ_E on all of E. In the basis $u = x_1 + x_2$, $v = x_1 - x_2$, t, the relations read $uv = vu$, $tu = ut$ and $tv = -vt$, so B is a skew polynomial ring and E is its classical point scheme [4]. The D_8-grading is not diagonal in this basis, since γ exchanges x_1 and x_2, and its identity component $B_e = \Bbbk[x_1^2, x_2^2, t^2]$ is a commutative polynomial ring. The computation agrees with the description of point schemes of quadratic Ore extensions of twists of polynomial rings by Lozano and Vancliff [32].

9. Homological determinants

In this section we compute homological determinants of graded Ore extensions and of identity components of dual reflection groups, using trace series.

Let R be a connected graded algebra and θ a graded automorphism of R. The *trace series* of θ is

$$\mathrm{Tr}_R(\theta, s) = \sum_{n \geq 0} \mathrm{tr}\left(\theta|_{R_n}\right) s^n.$$

For $\theta = \mathrm{id}$ this is the Hilbert series $H_R(s)$. We use the homological determinant $\mathrm{hdet}_R(\theta) \in \Bbbk^\times$ of Jørgensen and Zhang [22] through the following property. Let R be a noetherian AS regular algebra of dimension d with Gorenstein parameter ℓ, and θ a graded automorphism of R. Then θ is rational in the sense of [22, 1.3] by [22, 4.2], so $\mathrm{Tr}_R(\theta, s)$ is a rational function of s, and its Laurent expansion in s^{-1} has the form [22, Lemma 2.6]

$$\mathrm{Tr}_R(\theta, s) = (-1)^d \,\mathrm{hdet}_R(\theta)^{-1} s^{-\ell} + \sum_{j > \ell} a_j s^{-j}. \tag{9.1}$$

Since a rational function has a unique Laurent expansion in s^{-1}, (9.1) determines ℓ and $\mathrm{hdet}_R(\theta)$.

Proposition 9.2. *Let $S = R[t; \sigma, \delta]$ be a graded Ore extension with $\deg t = m$, with R noetherian and AS regular, and let θ be a graded automorphism of S with $\theta(R) = R$. Then $\theta(t) = \lambda t + a$ with $\lambda \in \Bbbk^\times$ and $a \in R_m$, and*

$$\mathrm{Tr}_S(\theta, s) = \frac{\mathrm{Tr}_R(\theta|_R, s)}{1 - \lambda s^m}, \quad \mathrm{hdet}_S(\theta) = \lambda \,\mathrm{hdet}_R(\theta|_R). \tag{9.3}$$

Proof. By Corollary 3.11, S is a noetherian AS regular algebra of dimension $d+1$ with Gorenstein parameter $\ell_R + m$, where $d = \dim R$. Since $S_m = R_m \oplus \Bbbk t$, we can write $\theta(t) = \lambda t + a$ with $a \in R_m$, and $\lambda \neq 0$ since otherwise $\theta(S) \subseteq R$. Consider the filtration of S by the graded subspaces

$$F_j S = \bigoplus_{i \leq j} R t^i.$$

By (2.6), $(F_i S)(F_j S) \subseteq F_{i+j} S$. Hence $(\lambda t + a)^i \in \lambda^i t^i + F_{i-1} S$, and $\theta(b t^i) = \theta(b)(\lambda t + a)^i \in \lambda^i \theta(b) t^i + F_{i-1} S$ for $b \in R$. So θ preserves the filtration and acts on $F_i S / F_{i-1} S \cong R(-mi)$ as $\lambda^i \theta|_R$. Since $S_n \subseteq F_j S$ for $mj \geq n$, we get

$$\mathrm{Tr}_S(\theta, s) = \sum_{i \geq 0} \lambda^i s^{mi} \,\mathrm{Tr}_R(\theta|_R, s) = \frac{\mathrm{Tr}_R(\theta|_R, s)}{1 - \lambda s^m}$$

as power series, hence as rational functions. The Laurent expansion in s^{-1} is multiplicative. Using (9.1) for R and

$$\frac{1}{1 - \lambda s^m} = -\lambda^{-1} s^{-m} \big(1 + \lambda^{-1} s^{-m} + \lambda^{-2} s^{-2m} + \cdots\big),$$

the leading term of the right side is $(-1)^{d+1} \big(\lambda \,\mathrm{hdet}_R(\theta|_R)\big)^{-1} s^{-(\ell_R + m)}$. Comparing with (9.1) for S gives (9.3). □

When $\delta = 0$ and $a = 0$, the element t is normal with $\theta(t) = \lambda t$, and Proposition 9.2 is a case of the normal-element formula of Jing and Zhang [21, Proposition 2.4], see also [25, Lemma 1.6]. For general δ and a, neither the σ-derivation nor the translation part a contributes to the homological determinant. Nakayama automorphisms of graded Ore extensions were computed by Liu, Wang and Wu [30, Theorem 3.3], by Shen and Guo for Koszul AS regular algebras [40], and by Liu and Ma for Ore extensions of polynomial algebras [31]. Combined with the theorem of Jørgensen and Zhang that invariants of automorphisms with trivial homological determinant are AS Gorenstein [22, Theorem 3.3], Proposition 9.2 gives the following criterion.

Corollary 9.4. *Let $S = R[t;\sigma,\delta]$ be as in Proposition 9.2, and let Θ be a finite group of graded automorphisms of S with $\theta(R) = R$ for every $\theta \in \Theta$. For $\theta \in \Theta$ write $\theta(t) = \lambda_\theta t + a_\theta$. If $\operatorname{hdet}_R(\theta|_R) = \lambda_\theta^{-1}$ for every $\theta \in \Theta$, then S^Θ is AS Gorenstein.*

Proof. By (9.3), $\operatorname{hdet}_S(\theta) = 1$ for every $\theta \in \Theta$, and [22, Theorem 3.3] applies. □

For example, let $S = \Bbbk_q[x,y] = \Bbbk[x][y;\sigma]$ and let $\zeta \in \Bbbk^\times$ be a root of unity of order N. The automorphism $\theta(x) = \zeta x$, $\theta(y) = \zeta^{-1}y$ generates a group of order N, with $\theta(\Bbbk[x]) = \Bbbk[x]$, $\lambda_\theta = \zeta^{-1}$ and $\operatorname{hdet}_{\Bbbk[x]}(\theta|_{\Bbbk[x]}) = \zeta = \lambda_\theta^{-1}$. So $\operatorname{hdet}_S(\theta) = 1$, and $S^{\langle\theta\rangle}$ is AS Gorenstein. For $q = 1$ this is the invariant ring $\Bbbk[x^N, xy, y^N]$ of the cyclic subgroup of SL_2 of order N.

Example 9.5. Proposition 9.2 applies when $\delta \neq 0$ and $a \neq 0$. Let $S = \Bbbk[x][y;\delta]$ be the Jordan plane, and let θ be given by

$$\theta(x) = \lambda x, \quad \theta(y) = \lambda y + \mu x \quad (\lambda \in \Bbbk^\times,\ \mu \in \Bbbk).$$

It preserves the relation, since $\theta(y)\theta(x) - \theta(x)\theta(y) - \theta(x)^2 = \lambda^2(yx - xy - x^2)$. Here $a = \mu x$, and θ acts on $\Bbbk[x]_n = \Bbbk x^n$ by λ^n. So

$$\operatorname{Tr}_S(\theta, s) = \frac{1}{(1-\lambda s)^2} = \lambda^{-2}s^{-2} + 2\lambda^{-3}s^{-3} + \cdots,$$

and $\operatorname{hdet}_S(\theta) = \lambda^2 = \lambda \operatorname{hdet}_{\Bbbk[x]}(\theta|_{\Bbbk[x]})$, as predicted by (9.3). The term μx does not contribute.

We relate the homological determinants of an algebra and of the identity component of a dual reflection group.

Proposition 9.6. *Let A be a noetherian AS regular domain generated in degree one, G a dual reflection group for A, and θ a graded automorphism of A with $\theta(A_g) = A_g$ for every $g \in G$. Write $\theta(f_g) = \lambda_g f_g$ with $\lambda_g \in \Bbbk^\times$. Then*

$$\operatorname{Tr}_A(\theta, s) = p_\theta(s)\ \operatorname{Tr}_{A_e}(\theta|_{A_e}, s), \quad p_\theta(s) = \sum_{g\in G} \lambda_g s^{\deg f_g}, \tag{9.7}$$

and

$$\operatorname{hdet}_{A_e}(\theta|_{A_e}) = \lambda_{m_G}\ \operatorname{hdet}_A(\theta), \quad \ell_{A_e} = \ell_A + N_G. \tag{9.8}$$

Proof. The scalars λ_g exist by Section 2. We have $\theta(A_e) = A_e$ and $\theta(f_g b) = \lambda_g f_g \theta(b)$ for $b \in A_e$. Since $A = \bigoplus_g f_g A_e$ and left multiplication by f_g is injective, this gives (9.7). The algebra A_e is AS regular by definition and noetherian by Section 2, and $\dim A_e = \dim A$ by [27, Lemma 3.4(1)], since A is free of finite rank over A_e on both sides. So (9.1) applies to θ and to $\theta|_{A_e}$, with the same sign $(-1)^d$. By (2.15), the leading term of p_θ is $\lambda_{m_G} s^{N_G}$. Comparing the leading terms of the Laurent expansions in s^{-1} of both sides of (9.7) gives

$$(-1)^d \operatorname{hdet}_A(\theta)^{-1} s^{-\ell_A} = (-1)^d \lambda_{m_G} \operatorname{hdet}_{A_e}(\theta|_{A_e})^{-1} s^{N_G - \ell_{A_e}},$$

which is (9.8). □

The identity $\ell_{A_e} = \ell_A + N_G$ is [27, Lemma 3.4(3)], since the Frobenius algebra $C(A)$ has Gorenstein parameter $-N_G$.

Example 9.9. In Example 2.22, let $\theta(x) = \alpha x$ and $\theta(y) = \beta y$. Then $\lambda_{m_G} = \lambda_\tau = \alpha$, and

$$\operatorname{Tr}_A(\theta, s) = \frac{1}{(1-\alpha s)(1-\beta s)}, \quad \operatorname{Tr}_{A_e}(\theta|_{A_e}, s) = \frac{1}{(1-\alpha^2 s^2)(1-\beta s)}, \quad p_\theta(s) = 1 + \alpha s.$$

The identity (9.7) holds since $(1+\alpha s)(1-\alpha s) = 1 - \alpha^2 s^2$. The leading terms of the expansions in s^{-1} are $(\alpha\beta)^{-1}s^{-2}$ and $(\alpha^2\beta)^{-1}s^{-3}$, so $\operatorname{hdet}_A(\theta) = \alpha\beta$ and $\operatorname{hdet}_{A_e}(\theta|_{A_e}) = \alpha^2\beta$, as predicted by (9.8). Also $\ell_{A_e} = 3 = \ell_A + \deg f_\tau$.

For iterated Ore extensions, Propositions 9.2 and 9.6 give closed formulas.

Corollary 9.10. *Let B be the iterated Ore extension* (1.5)*, and let θ be a graded automorphism of B with $\theta(B^{(i)}) = B^{(i)}$ for every i. Then $\theta(x_i) = \lambda_i x_i + a_i$ with $\lambda_i \in \Bbbk^\times$ and $a_i \in (B^{(i-1)})_1$, and*

$$\operatorname{hdet}_B(\theta) = \lambda_1\lambda_2\cdots\lambda_n. \tag{9.11}$$

If moreover G is a dual reflection group for B, with every x_i homogeneous, and $\theta(B_g) = B_g$ for every $g \in G$, then

$$\theta(f_{m_G}) = \lambda_1^{r_1-1}\cdots\lambda_n^{r_n-1} f_{m_G}, \quad \operatorname{hdet}_{B_e}(\theta|_{B_e}) = \lambda_1^{r_1}\lambda_2^{r_2}\cdots\lambda_n^{r_n}. \tag{9.12}$$

Proof. Since $(B^{(i)})_1 = (B^{(i-1)})_1 \oplus \Bbbk x_i$ and θ preserves $B^{(i)}$ and $B^{(i-1)}$, we have $\theta(x_i) = \lambda_i x_i + a_i$ with $a_i \in (B^{(i-1)})_1$, and $\lambda_i \neq 0$ since $\theta(B^{(i)}) = B^{(i)}$. By Proposition 9.2 applied to $B^{(i)} = B^{(i-1)}[x_i; \sigma_i, \delta_i]$, we get $\operatorname{hdet}_{B^{(i)}}(\theta) = \lambda_i \operatorname{hdet}_{B^{(i-1)}}(\theta)$, and (9.11) follows by induction, since $\operatorname{hdet}_{\Bbbk}(\theta) = 1$.

For the second statement, put $g_i = x_1^{r_1-1} \cdots x_i^{r_i-1}$, so that $f_{m_G} = g_n$ by Theorem C, and $\theta(g_n) = \lambda_{m_G} g_n$. Expanding $(\lambda_n x_n + a_n)^{r_n-1} = \lambda_n^{r_n-1} x_n^{r_n-1} + (\text{lower powers of } x_n)$ and comparing the coefficients of $x_n^{r_n-1}$ in the free left $B^{(n-1)}$-module B, we get $\theta(g_{n-1})\lambda_n^{r_n-1} = \lambda_{m_G} g_{n-1}$. So g_{n-1} is an eigenvector of θ. Since $g_{n-1} = f_{m_{K_{n-1}}}$ for the dual reflection group K_{n-1} of $B^{(n-1)}$, the same argument and induction give $\lambda_{m_G} = \prod_i \lambda_i^{r_i-1}$. Now Proposition 9.6 gives $\operatorname{hdet}_{B_e}(\theta|_{B_e}) = \lambda_{m_G} \operatorname{hdet}_B(\theta)$, and (9.11) gives (9.12). □

The formula (9.12) can also be obtained from Proposition 5.6. The automorphism θ preserves every $B_e^{(i)}$, and $\theta(x_i^{r_i}) = (\lambda_i x_i + a_i)^{r_i}$ has leading coefficient $\lambda_i^{r_i}$ in the variable $y_i = x_i^{r_i}$ of the iterated Ore extension (5.7), so Proposition 9.2, applied to each step of (5.7), gives the same value.

Example 9.13. In Example 5.13, let $\mu, \lambda \in \Bbbk^\times$. The assignment $\theta(x_i) = \mu x_i$ and $\theta(t) = \lambda t$ defines a graded automorphism of B, since $\theta(tx_i - x_{i+1}t) = \lambda\mu(tx_i - x_{i+1}t)$. It preserves every $B^{(i)}$ and every component B_g, since it multiplies each monomial by a scalar. By Corollary 9.10,

$$\operatorname{hdet}_B(\theta) = \lambda\mu^n, \quad \theta(f_{m_G}) = \mu^{n(r-1)}\lambda^{np-1} f_{m_G}, \quad \operatorname{hdet}_{B_e}(\theta|_{B_e}) = \mu^{nr}\lambda^{np}.$$

This agrees with the direct computation in the polynomial ring $B_e = \Bbbk[x_1^r, \ldots, x_n^r, t^{np}]$, on which θ acts by μ^r on each x_i^r and by λ^{np} on t^{np}. If θ has finite order and $\lambda\mu^n = 1$, then $B^{\langle\theta\rangle}$ is AS Gorenstein by Corollary 9.4. For $n = r = 2$ and $p = 1$, the table of Example 5.17 gives $p_\theta(s) = (1+\mu s)^2(1+\lambda s)$, and

$$\operatorname{Tr}_B(\theta, s) = \frac{1}{(1-\mu s)^2(1-\lambda s)}, \quad \operatorname{Tr}_{B_e}(\theta|_{B_e}, s) = \frac{1}{(1-\mu^2 s^2)^2(1-\lambda^2 s^2)}.$$

The leading terms of the expansions in s^{-1} are $-(\mu^2\lambda)^{-1}s^{-3}$ and $-(\mu^4\lambda^2)^{-1}s^{-6}$, so $\operatorname{hdet}_B(\theta) = \lambda\mu^2$ and $\operatorname{hdet}_{B_e}(\theta|_{B_e}) = \lambda^2\mu^4 = \lambda_{m_G} \operatorname{hdet}_B(\theta)$ with $\lambda_{m_G} = \mu^2\lambda$, and $\ell_{B_e} = 6$.

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